CCDE Study Guide

Marwan Al-shawi
CCDE No. 20130066

Cisco Press

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Warning and Disclaimer

This book covers various possible design options available while working across multiple places in the network. As part of the evaluation process, the book stays focused on various technical requirements, business requirements, constraints, and associated implications rather than on providing best practice recommendations.

Every effort has been made to make this book as comprehensive and as accurate as possible. The book does not attempt to teach foundational knowledge. Please supplement your learning and fill in gaps in knowledge by reviewing separate technology-specific publications as part of your journey to become a Design Expert.

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Dedication

I would like to dedicate this book to my wonderful mother for her continued support, love, encouragement, guidance, and wisdom.

And most importantly, I would like to thank God for all the blessings in my life.

—Marwan

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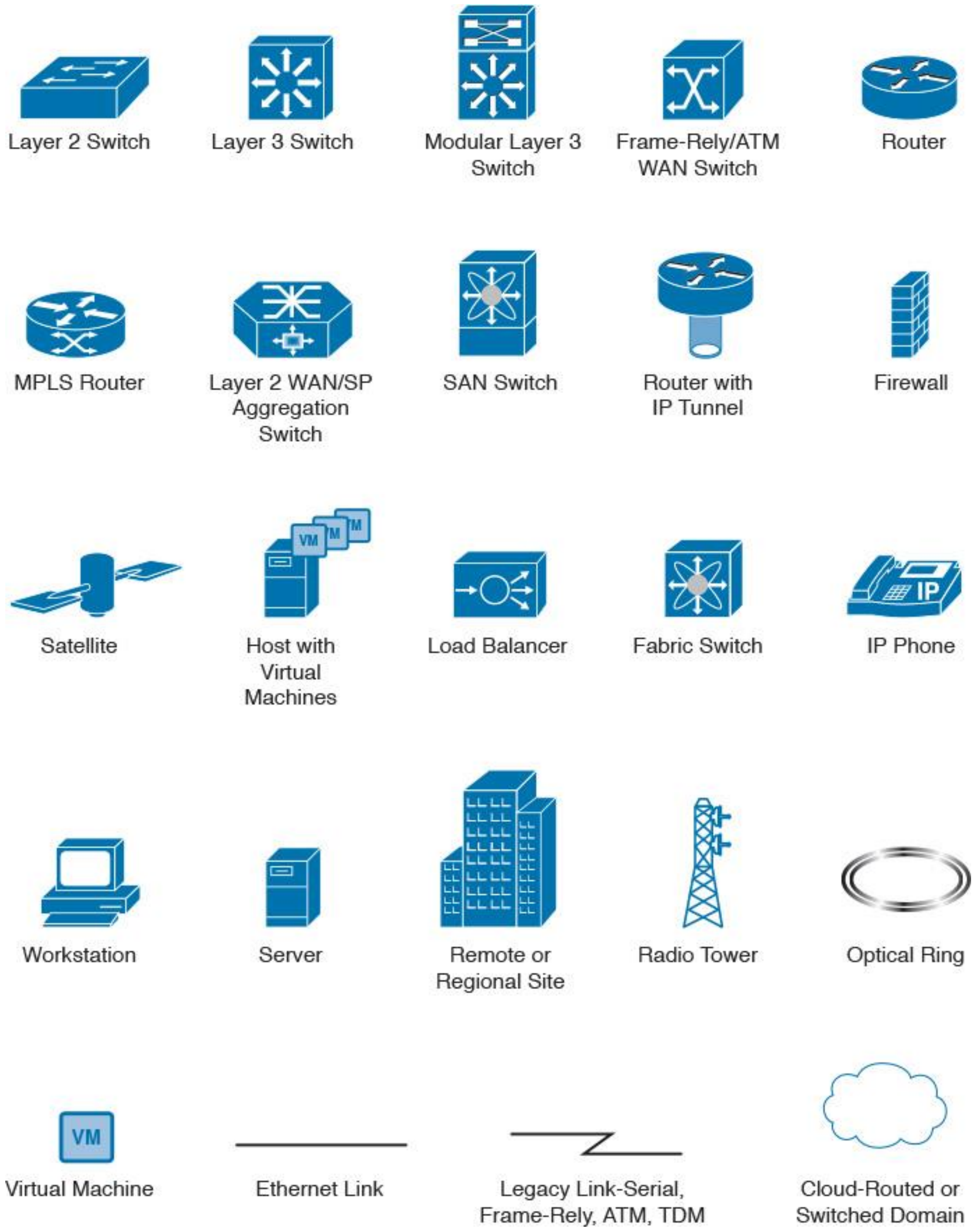
Contents at a Glance

Contents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Firewall
Satellite
Host with Virtual Machines
Load Balancer
Fabric Switch
IP Phone
Workstation
Server
Remote or Regional Site
Radio Tower
Optical Ring
Virtual Machine
Ethernet Link
Legacy Link-Serial, Frame-Rely, ATM, TDM
Cloud-Routed or Switched Domain

Introduction

The CCDE certification is a unique certification in the IT and networking industry and is considered one of the most if not the only recognized vendor-neutral network Design Expert level certification. When it comes to design, it is like art: It cannot be taught or covered entirely through a single book or a training course, because each design has different drivers, philosophy, and circumstances that collectivity create its unique characteristic. Therefore, this book uses a comparative and analytical approach to help the reader answer the question why with regard to design or technology selections, and to think of how to link the technical design decisions to other influencing factors (technical, nontechnical, or combination of both) to achieve a true and successful business-driven design. In addition, multiple mini design scenarios and illustrations are included in the chapters to explain the concepts, design options, and implications.

This book is the first book to target the CCDE practical exam. Also, It is the first book that consists of diverse design aspects, principles, and options using a business-driven approach for enterprise, service provider, and data center networks.

This book covers the different design principles and topics using the following approach:

- Covers (that is, highlights, discusses, and compares) the different technologies, protocols, design principles, and design options in terms of cost, availability, scalability, performance, flexibility, and so on (along with the strength of the various designs and design concerns where applicable)
- Covers the drivers toward adopting the different technologies and protocols (technical and nontechnical) (whether intended for enterprise or service provider networks depends on the topic and technology)
- Covers the implications of the addition or integration of any element to the overall design, such as adding new applications or integrating two different networks

The design topics covered in this CCDE Study Guide aim to prepare you to be able to

- Analyze and identify various design requirements (business, functional, and application) that can influence design decisions.
- Understand the different design principles and their impact on the organization from a business point of view
- Understand and compare the various network design architectures and the associated implications on various design aspects of applying different Layer 2 and Layer 3 control plane protocols
- Identify and analyze design limitations or issues, and how to optimize them, taking into consideration the technical and nontechnical design requirements and constraints.
- Identify and analyze the implication of adding new services or applications and how to accommodate the design or the design approach to meet the expectations

This book references myriad sources, but presents the material in a manner tailored for conscientious network designers and architects. The material also covers updated standards and features that are found in enterprise and service provider networks. In addition, each chapter contains further reading suggestions pointing you to recommended documents that pertain to the topics covered in each chapter.

Therefore, you can use this book as an all-in-one study guide covering the various networking technologies, protocols, and design options in a business-driven approach. You can expand your study scope and depth of knowledge selectively on certain topics as needed.

Whether you are preparing for the CCDE certification or just want to better understand advanced network design topics, you will benefit from the range of topics covered and the practical business-driven approach used to analyze, compare, and explain these topics.

Who Should Read This Book?

In addition to those who are planning or studying for the CCDE certification, this book is for network engineers, network consultants, network architects, and solutions architects who already have a foundational knowledge of the topics being covered and who would like to train themselves to think more like a design engineer rather than like an implementation engineer.

CCDE Practical Exam Overview

The minimally qualified CCDE must have expert-level knowledge, experience, and skills that cover complex networks design (ideally global-scale networks) by successfully demonstrating the ability to translate business requirements and strategies into functional and technical strategies. In other words, CCDEs are recognized for their expert-level knowledge and skills in network infrastructure design. The deep technical networking knowledge that a CCDE brings ensures that they are well qualified to address the most technically challenging network infrastructure design assignments [1]. Therefore, to test the CCDE candidate skills, knowledge, and expertise, the CCDE practical exam is divided into multiple design scenarios, with each having a different type of network and requirements. In addition, each design scenario is structured of different domains and tasks that CCDE candidates have to complete to demonstrate expert-level abilities when dealing with a full network design lifecycle (gather business requirements, analyze the requirements, develop a design, plan the implementation of the design, and then apply and optimize the design).

Job Tasks

The CCDE exam is designed to cover different use cases, each of which may be integrated in one or multiple design scenarios in the exam. The following are the primary use cases at the time of this writing:

- **Merge or divest networks:** This use case covers the implications and challenges (technical and nontechnical) associated with merging or separating different networks (both enterprise and service provider types of networks). This domain, in particular, can be one of the most challenging use cases for network designers because, most of the time, merging two different networks means integrating two different design philosophies, in which multiple conflicting design concepts can appear. Therefore, at a certain stage, network designers have to bring these two different networks together to work as a one cohesive system, taking into consideration the various design constrains that might exist such as goals for the merged network, security policy compliance, timeframe, cost, the merger constraints, the decision of which services to keep and which ones to divest (and how), how to keep services up and running after the divestiture, what the success criteria is for the merged network, and who is the decision maker.

- **Add technology or application:** This use case covers the impact of adding technology or an application to an existing network. Will anything break as a result of the new addition? In this use case, you must consider the application requirements in terms of traffic pattern, convergence time, delay, and so on across the network. By understanding these requirements, the CCDE candidate as a network designer should be able to make design decisions about fine-tuning the network (design and features such as quality of service [QoS]) to deliver this application with the desired level of experience for the end users.

- **Replace technology:** This use case covers a wide range of options to replace an existing technology to meet certain requirements. It might be a WAN technology, routing protocol, security mechanism, underlying network core technology, or so on. Also, the implications of this new technology or protocol on the current design, such as enhanced scalability or potential to conflict with some of the existing application requirements, require network designers to tailor the network design so that these technologies work together rather than in isolation, so as to reach objectives, such as delivering business applications and services.

- **Scale a network:** This use case covers different types of scalability aspects at different levels, such as physical topology, along with Layer 2 and Layer 3 scalability design considerations. In addition, the challenges associated with the design optimization of an existing network design to offer a higher level of scalability are important issues in this domain. For example, there might be some constraints or specific business requirements that might limit the available options for the network designer when trying to optimize the current design. Considerations with regard to this use case include the following: Is the growth planned or organic? Are there issues caused by the growth? Should one stop and redesign the network to account for growth? What is the most scalable design model?

Exam Job Domains

In each of the CCDE exam use cases described in the preceding section, as part of each CCDE design scenario the candidate is expected to cover some or all of the following job domains:

- **Analyze:** This domain requires identification of the requirements, constraints, and risks from both business and technology perspectives. In this task, the candidate is expected to perform multiple subtasks, such as the following:
- Identify the missing information required to produce a design.
- Integrate and analyze information from multiple sources (for example, from e-mails or from diagrams) to provide the correct answer for any given question.
- **Design:** In this domain, the CCDE candidate is usually expected to provide a suggested design by making design choices and decisions based on the given and analyzed information and requirements in the previous task. Furthermore, one of the realistic and unique aspects of the CCDE exam is that there might be more than one right design option. Also, there might be optimal and suboptimal solutions. This aspect of the exam is based on the CCDE candidate's understanding of the requirements, goals, and constraints in making the most relevant and suitable selection given the options available.
- **Implement and deploy:** This domain contains multiple subtasks, such as the following:
- Determine the consequences of the implementation of the proposed design.
- Design implementation, migration, or fallback plans.

Note

No command-line interface (CLI) configuration is required on the CCDE practical exam. The general goal behind this point is more about how and where you to apply a network technology or a protocol and the implications associated with it, and how to generate a plan to apply the proposed design.

- **Validate and optimize:** Here the CCDE candidate is required to justify the design choices and decisions in terms of the rationale behind the design's selection. The candidate's justifications should evidence that the selected design is the best available. Justifications are typically driven by technical requirements, business requirements, and functional requirements, considered either in isolation or in combination.

Exam Technologies

As a general rule for the CCDE practical exam technologies, consider the written exam (qualification) blueprint as a reference (see Figure I-1). However, remember that this is a scenario-based design exam, in which you might expect expansion to the technologies covered in the CCDE written blueprint. In other words, consider the blueprint as a foundation and expand upon it to a reasonable extent; it is not necessary to go deeply into technologies that are not used in real-life networks.

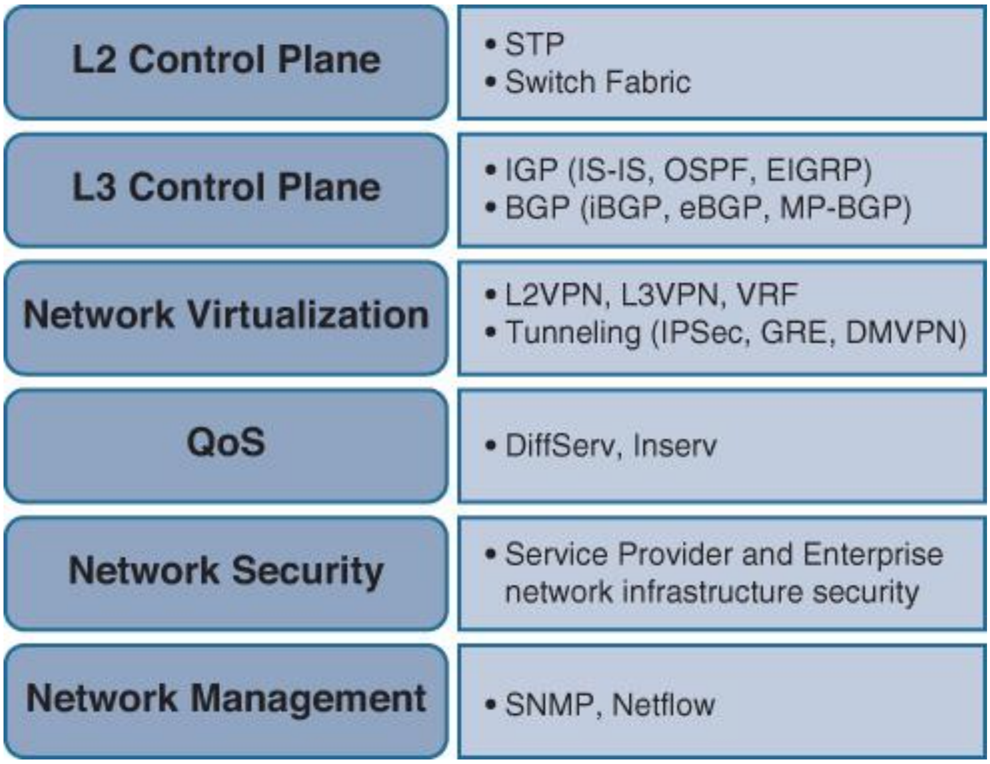

Figure I-1 *Exam Technologies*

Note

The above technologies include both IP versions (version 4 and 6) as well as unicast and Multicast IP communications.

PPDIOO Approach and the CCDE Job Domains

With regard to IT services, businesses usually aim to reduce total cost of ownership, improve its service availability, enhance user quality of experience, and reduce operational expenses. By adopting a lifecycle approach, organizations can define a set of methodologies and standards to be followed at each stage of the IT network's life. With this approach, there will be a series of phases that all collectively construct a lifecycle. With most lifecycle approaches, the information and findings of each phase can be used to feed and improve the following phase. This ultimately can produce more efficient and cost-effective IT network solutions that offer IT more elasticity to enhance current investments and to adopt new IT technologies and justify their investment cost.

The PPDIOO lifecycle (see Figure I-2) stands for Prepare, Plan, Design, Implement, Operate, and Optimize, which is Cisco's vision of the IT network lifecycle. This vision is primarily based on the concept that understanding what is supposed to happen at each stage is vital for a company (or architect, designer) to properly use the lifecycle approach and to get the most benefit from it.

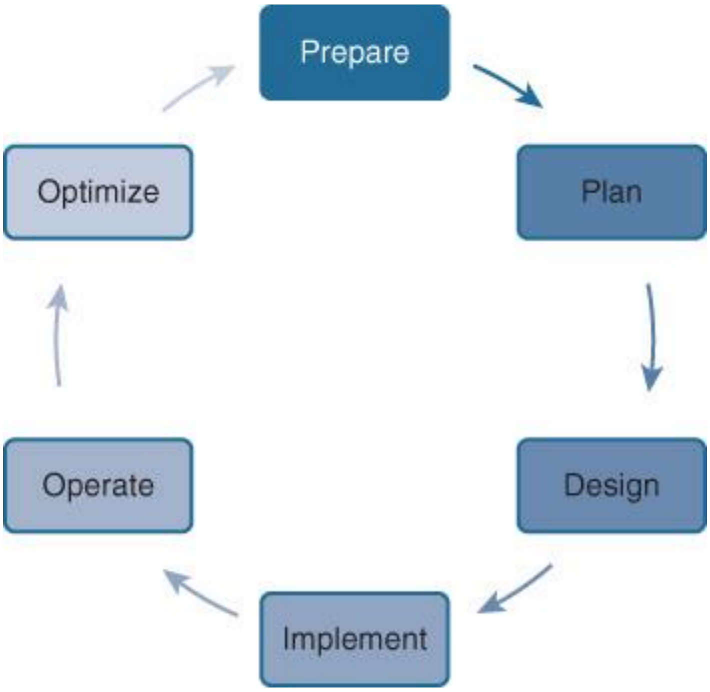

Figure I-2 *PPDIOO Lifecycle*

Furthermore, this approach offers the flexibility to have a two-way directional relationship between the phases. For instance, during the monitoring phase of the newly designed and implemented network, issues might be discovered that can be fixed by the addition of some features. This is similar to when there are issues related to design limitations. Therefore, each phase can provide reverse feeding, as well, to the previous phase or phases to overcome issues and limitations that appear during the project lifecycle. As a result, this will provide an added flexibility to IT in general and the design process in particular to provide a workable design that can transform the IT network infrastructure to be a business enabler. Figure I-3 illustrates the PIDDO lifecycle with the multidimensional relationship between the lifecycle phases.

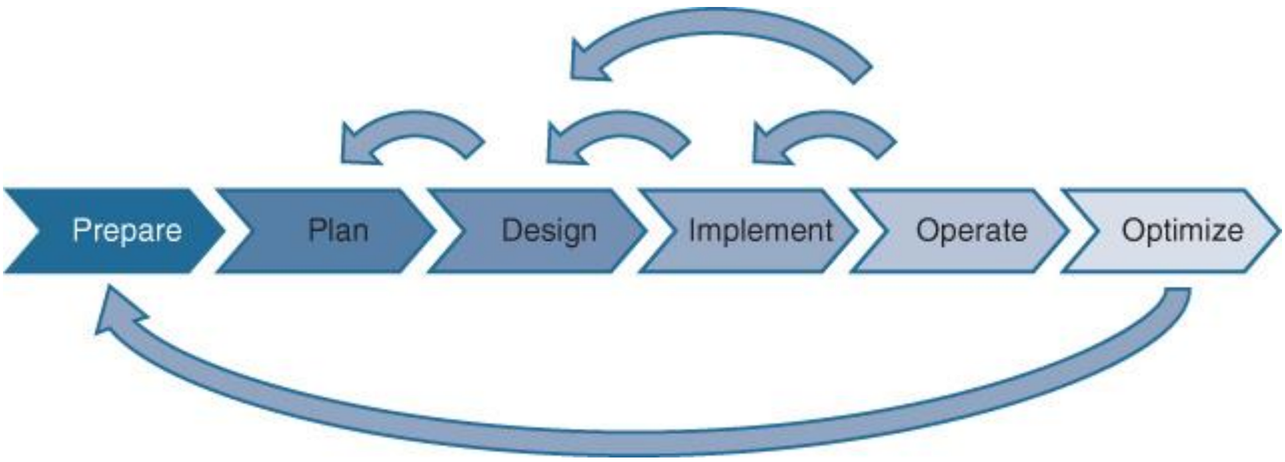

Figure I-3 *PPDIOO Multidimensional Relationship*

PPDIOO and Tasks

In fact, the PPDIOO lifecycle is applicable to the CCDE job domains, just like any other design project:

- The CCDE candidate needs to analyze the provided information (Prepare).
- Use this information to make design choices and decisions (Plan).
- Generate, propose, or suggest a suitable design (Design).
- Apply the selected design (for example, an implementation plan) (Implement).
- Collect feedback or monitor (Operate) for optimization and enhancement (Optimize).

Final Thoughts on the CCDE Practical Exam

Understanding the various domains and tasks expected in each of the exam's design scenarios can help CCDE practical exam candidates shape their study plans. This understanding can also help those who have the opportunity to practice it in their work environment. If they are working on a design and architecture project, they will have a tangible practical feeling and understand how the different stages of the design process can be approached and handled.

How This Book Is Organized

Although this book could be read cover to cover, it is designed to be flexible and allow you to easily move between chapters and sections of chapters to cover just the topics that you need more work with.

This book is organized into six distinct sections.

Part I of the book explains briefly the various design approaches, requirements, and principles, and how they complement each other to achieve a true business-driven design.

- **Chapter 1, "Network Design Requirements: Analysis and Design Principles"** This chapter covers how the different requirements (business, functional, and application) can influence design decisions and technology selection to achieve a business-driven design. This chapter also examines, when applicable, the foundational design principles that network designers need to consider.

Part II of this book focuses on the enterprise networks, specifically modern (converged) networks. The chapter covers the various design options, considerations, and design implications with regard to the business and other design requirements.

- **Chapter 2, "Enterprise Layer 2 and Layer 3 Design"** This chapter covers different design options and considerations related to Layer 2 and Layer 3 control plane protocols and advanced routing concepts.
- **Chapter 3, "Enterprise Campus Architecture Design"** This chapter covers the design options applicable to modern campus networks. The chapter also covers some of the design options and considerations with regard to network virtualization across the campus network.
- **Chapter 4, "Enterprise Edge Architecture Design"** This chapter covers various design options and considerations with regard to the two primary enterprise edge blocks (WAN edge and Internet edge).

Part III of the book focuses on service provider-grade networks. It covers the various design architectures, technologies, and control protocols, along with the drivers toward adopting the different technologies (technical and nontechnical).

- **Chapter 5, "Service Provider Network Architecture Design"** This chapter covers the various architectural elements that collectively comprise a service provider-grade network at different layers (topological and protocols layers). The chapter also covers the implications of some technical design decisions on the business.
- **Chapter 6, "Service Provider MPLS VPN Services Design"** This chapter covers various options and considerations in MPLS VPN network environments, focusing on L2VPN and L3VPN networks. The chapter also examines different design options and approaches to optimize Layer 3 control plane design scalability for service provider-grade networks.
- **Chapter 7, "Multi-AS Service Provider Network Design"** This chapter focuses on the design options and considerations when interconnecting different networks or routing domains. The chapter examines each design option and then compares them based on various design aspects such as security and scalability.

Part IV of the book focuses on data center networks design for both traditional and modern (virtualized and cloud based) data center networks. This part also covers how to achieve business continuity goals.

- **Chapter 8, "Data Center Networks Design"** This chapter covers various design architectures, concepts, techniques, and protocols that pertain to traditional and modern data center networks. In addition, this chapter analyzes and compares the different design options and considerations, and examines the associated implications of interconnecting dispersed data center networks and how these different technologies and design techniques can facilitate achieving different levels of business continuity.

Part V of this book focuses on the design principles and aspects to achieve the desired levels of operational uptime and resiliency by the business.

- **Chapter 9, "Network High-Availability Design"** This chapter covers the different variables and factors that either solely or collectively influence the overall targeted operational uptime. This chapter also examines the various elements that influence achieving the desired degree of network resiliency and fast convergence.

Part VI of the book focuses on network technologies and services that are not core components of the CCDE practical exam.

- **Chapter 10, "Design of Other Network Technologies and Services"** This chapter briefly explains some design considerations with regard to the following network technologies and services, with a focus on certain design aspects and principles and without going into deep technical detail or explanation: IPv6, multicast, QoS, security, and network management.

Final Words

This book is an excellent self-study resource to learn how to think like a network designer following a business-driven approach. You will learn how to analyze and compare different design options, principles, and protocols based on various design requirements. However, the technical knowledge forms only the foundation to pass the CCDE practical exam. You also want to have a real feeling for gathering business requirements, analyzing the collected information, identifying the gaps, and producing a proposed design or design optimization based on that information. If you believe that any topic in this book is not covered in enough detail, I encourage you to expand your study scope on that topic by using the recommended readings in this book and by using the recommended CCDE study resources available online atLearning@Cisco.com

Enjoy the reading and happy learning.

Part I: Business-Driven Strategic Network Design

Chapter 1 Network Design Requirements: Analysis and Design Principles

Chapter 1. Network Design Requirements: Analysis and Design Principles

Designing large-scale networks to meet today's dynamic business and IT needs and trends is a complex assignment, whether it is an enterprise or service provider type of network. This is especially true when the network was designed for technologies and requirements relevant years ago and the business decides to adopt new IT technologies to facilitate the achievement of its goals but the business's existing network was not designed to address these new technologies' requirements. Therefore, to achieve the desired goal of a given design, the network designer must adopt an approach that tackles the design in a structured manner.

There are two common approaches to analyze and design networks:

- **The top-down approach:** The top-down design approach simplifies the design process by splitting the design tasks to make it more focused on the design scope and performed in a more controlled manner, which can ultimately help network designers to view network design solutions from a business-driven approach.
- **The bottom-up approach:** In contrast, the bottom-up approach focuses on selecting network technologies and design models first. This can impose a high potential for design failures, because the network will not meet the business or applications' requirements.

To achieve a successful strategic design, there must be additional emphasis on a business driven approach. This implies a primary focus on business goals and technical objectives, in addition to existing and future services and applications. In fact, in today's networks, business requirements are driving IT and network initiatives as shown in Figure 1-1 [6].

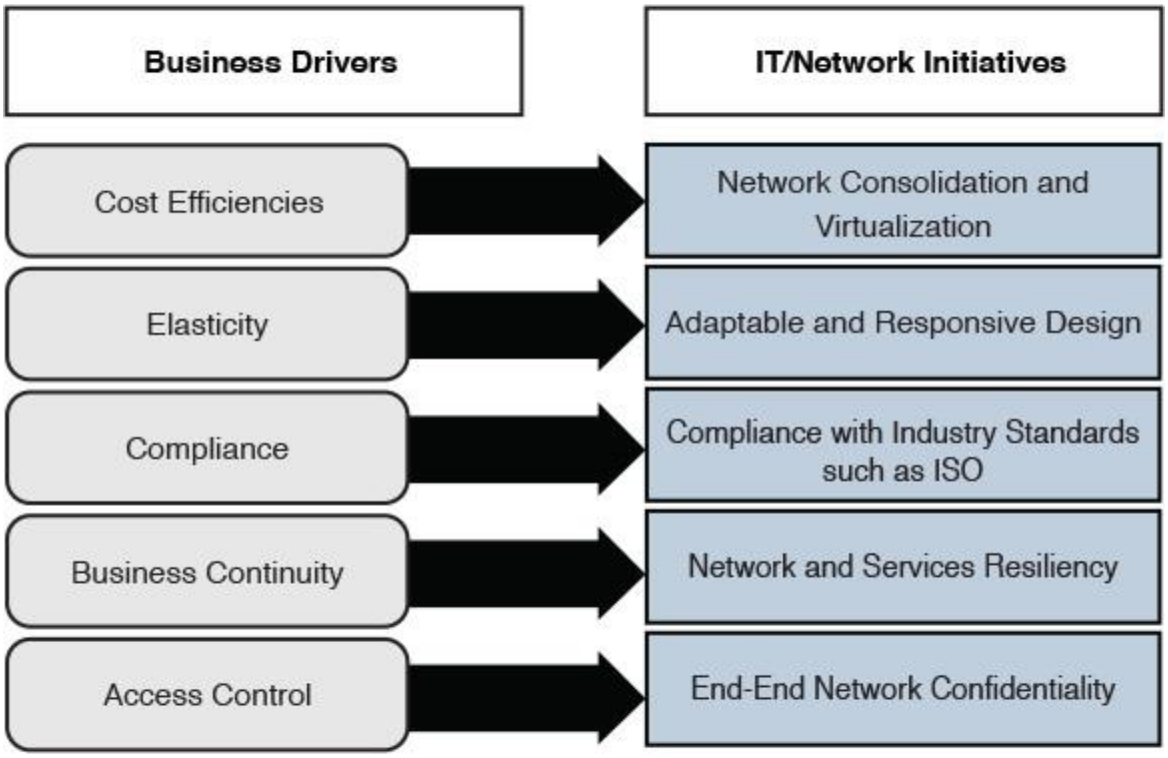

Figure 1-1 *Business Drivers Versus IT Initiatives*

For instance, although compliance (as presented in Figure 1-1) might seem to be a design constraint rather than a driver, many organizations today aim to comply with some standards with regard to their IT infrastructure and services to gain some business advantages, such as compliance with ISO/IEC 27001 Information Security Management,[1] will help businesses like financial services organizations to demonstrate their credibility and trust. This ultimately will help these organizations to gain more competitive advantages, optimize their operational uptime, and reduce operational expenses (fewer number of incidents as a result of the reduced number of information security breaches).

1. http://www.iso.org/iso/home/standards/management-standards/iso27001.htm

Throughout this book and for the purpose of the CCDE exam, the top-down approach is considered as the design approach that can employ the following top-down logic combined with the prepare, plan, design, implement, operate and optimize (PPDIOO) lifecycle:

- Analyze the goals, plans, and requirements of the business.
- Define application requirements from the upper layers of the Open Systems Interconnection (OSI) reference model that can help to identify the characteristics of an application.
- Specify the design of the infrastructure along with the functional requirements of its components, for the network to become a business enabler.
- Monitor and gather additional information that may help to optimize and influence the logical or physical design to adapt with any new application or requirements.

Design Scope

It is important in any design project that network designers carefully analyze and evaluate the scope of the design before starting to gather information and plan network design. Therefore, it is critical to determine whether the design task is for a green field (new) network or for a current production network (if the network already exists, the design tasks can vary such as optimization, expansion, integration with other external networks, and so on). It is also vital to determine whether the design spans a single network module or multiple modules. In other words, the predetermination of the design scope can influence the type of information required to be gathered, in addition to the time to produce the design.Table 1-1 shows an example of how identifying the design scope can help network designers determine the areas and functions a certain design must emphasize and address. As a result, the scope of the information to be obtained will more be focused on these areas.

Design Scope	Detailed Design Scope Example
Enterprise campus network and remote sites	Rollout of IP telephony across the enterprise, which may require a redesign of virtual LANs (VLANs), quality of service (QoS), and so on across the LAN, WAN, data center (DC), and remote-access edge
Campus only	Add multi-tenancy concept to the campus, which requires design of VLANs, IPs, and path isolation across the campus LAN only
Optimize enterprise edge availability	Add redundant link for remote access, which might require redesign of the WAN module and remote site designs and configurations such as overlay tunnels

Table 1-1 *Design Scope*

Note

Identifying the design scope in the CCDE exam is very important. For example, the candidate might have a large network to deal with, whereas the actual design focus is only on adding and integrating a new data center. Therefore, the candidate needs to focus on that part only. However, the design still needs to consider the network as a whole, a “holistic approach,” when you add, remove, or change anything across the network (as discussed in more detail later in this chapter).

Business Requirements

This section covers the primary aspects that pertain to the business drivers, needs, and directions that (individually or collectively) can influence design decisions either directly or indirectly. The best place to start understanding the business’s needs and requirements is by looking at the big picture of a company or business and understanding its goals, vision, and future directions. This can significantly help to steer the design to be more business driven. However, there can be various business drivers and requirements based on the business type and many other variables. As outlined inFigure 1-2, with a top-down design approach, it is almost always the requirements and drivers at higher layers (such as business and application requirements) that drive and set the requirements and directions for the lower layers. Therefore, network designers aiming to achieve a business-driven design must consider this when planning and producing a new network design or when evaluating and optimizing an existing one. The following sections discuss some of the business requirements and drivers at the higher layers and how each can influence design decisions at the lower layers.

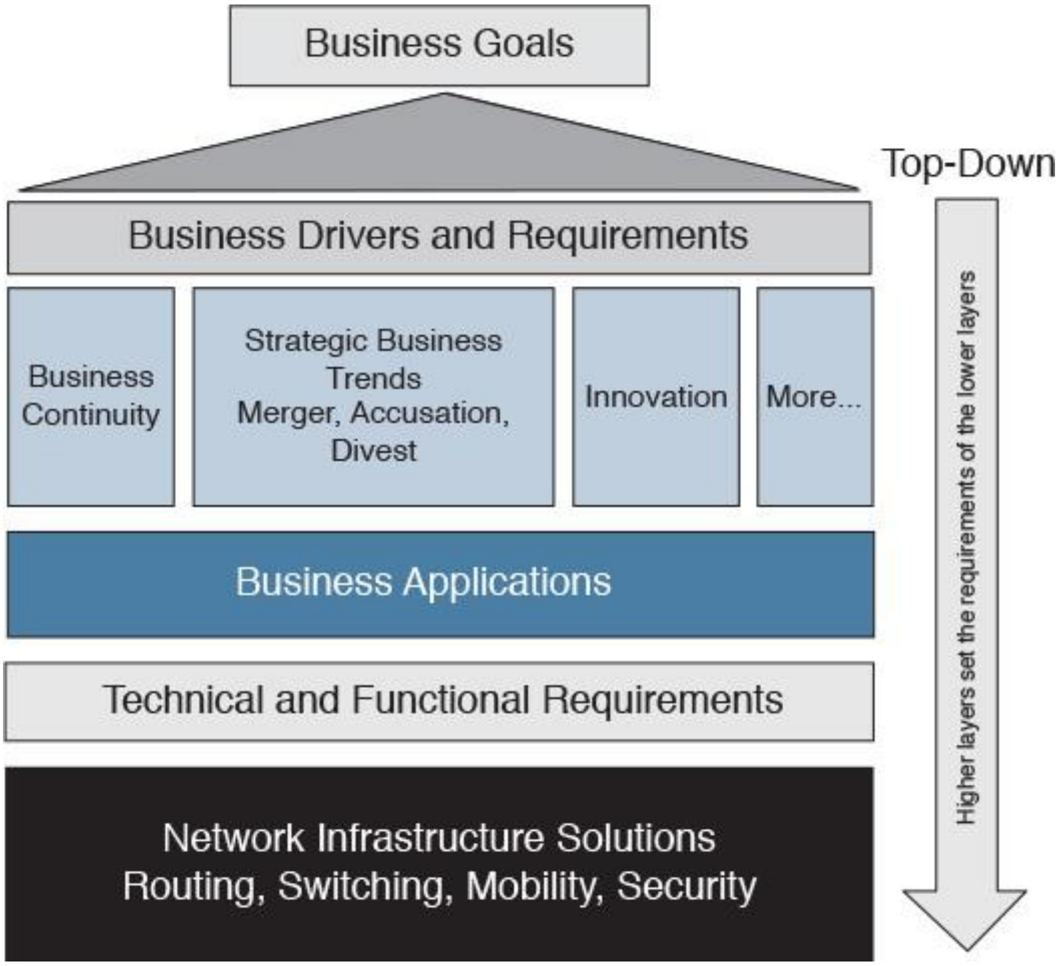

Figure 1-2 *Business-Driven Technology Solutions*

Business Continuity

Business continuity (BC) refers to the ability to continue business activities (business as usual) following an outage, which might result from a system outage or a natural disaster like a fire that damages a data center. Therefore, businesses need a mechanism or approach to build and improve the level of resiliency to react and recover from unplanned outages.

The level of resiliency is not necessarily required to be the same across the entire network, however, because the drivers of BC for the different parts of the network can vary based on different levels of impact on the business. These business drivers may include compliance with regulations or the level of criticality to the business in case of any system or site connectivity outage. For instance, if a retail business has an outage in one of its remote stores, this is of less concern than an outage to the primary data center, from a business point of view. If the primary data center were to go offline for a certain period of time, this would affect all the other stores (higher risk) and could cost the business a larger loss in terms of money (tangible) and reputation (intangible). Therefore, the resiliency of the data center network is of greater consideration for this retailer than the resiliency of remote sites [17].

Similarly, the location of the outage sometimes influences the level of criticality and design consideration. Using the same example, an outage at one of the small stores in a remote area might not be as critical as an outage in one of the large stores in a large city [11]. In other words, BC considerations based on risk assessment and its impact on the business can be considered one of the primary drivers for many businesses to adapt network technologies and design principles to meet their desired goals [5].

Elasticity to Support the Strategic Business Trends

Elasticity refers to the level of flexibility a certain design can provide in response to business changes. A change here refers to the direction the business is heading, which can take different forms. For example, this change may be a typical organic business growth, a decline in business, a merger, or an acquisition. For instance, if an enterprise campus has three buildings and is interconnected directly, as illustrated in Figure 1-3, any organic growth in this network that requires the addition of a new building to this network will introduce a lot of complexity in terms cabling, control plane, and manageability. These complexities result from the inflexible design, which makes the design incapable of responding to the business's natural growth demand.

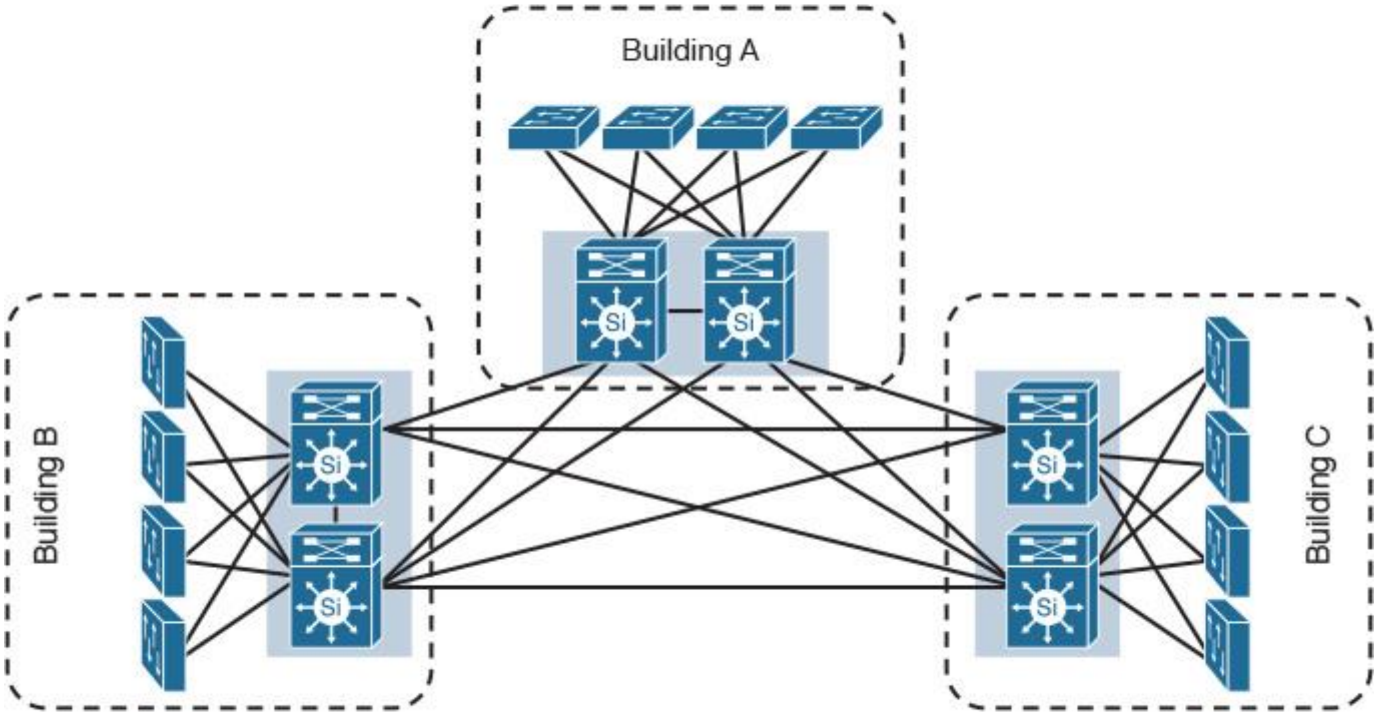

Figure 1-3 *Inflexible Design*

To enhance the level of flexibility of this design, you can add a core module to optimize the overall design modularity to support business expansion requirements. As a result, adding or removing any module or building to this network will not affect other modules, and does not even require any change to the other modules, as illustrated in Figure 1-4. In other words, the design must be flexible enough to support the business requirements and strategic goals. If network designers understand business trends and directions in this area, such understanding may influence, to a large extent, deign choices.

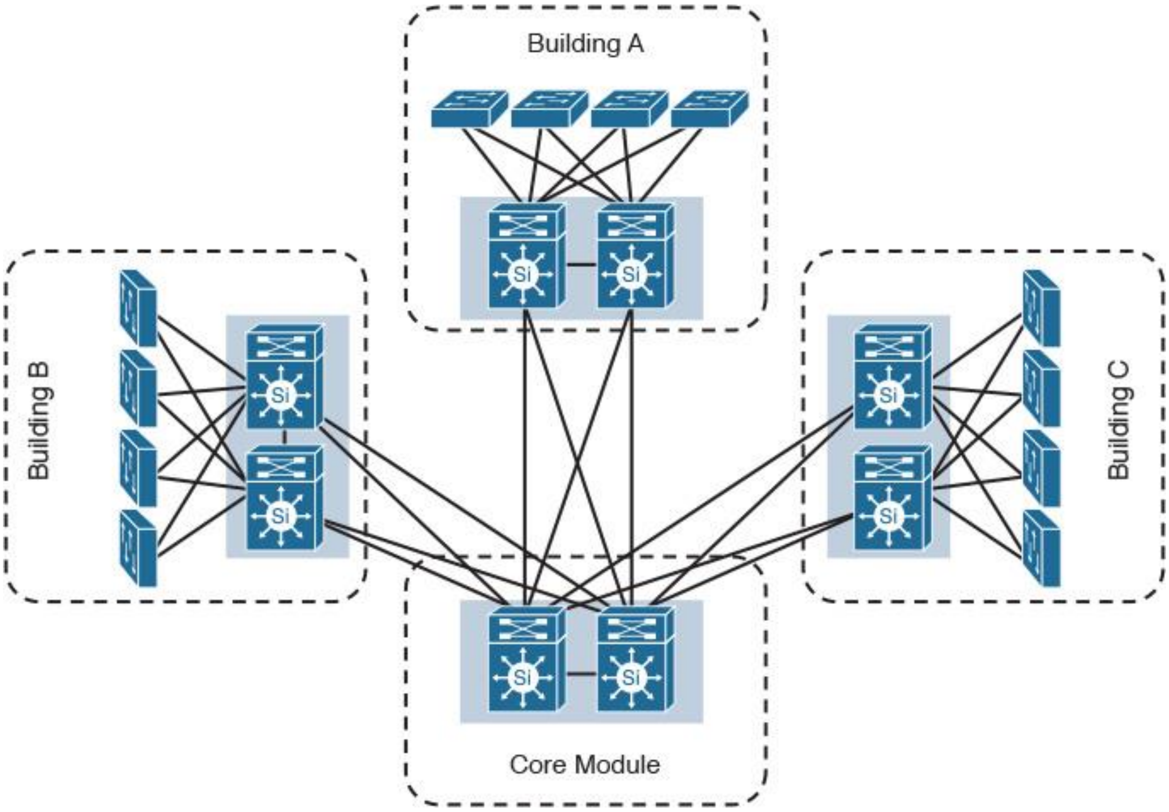

Figure 1-4 *Flexible Design*

Similarly, a flexible network design must support the capability to integrate with other networks (for examples, when mergers and acquisitions occur). With mergers and acquisitions, however, the network can typically grow significantly in size within a short period of time, and the biggest challenge, in both scenarios (mergers and acquisitions), is that network designers have to deal with different design principles, the possibility of overlapping IP address space, different control plane protocols, different approaches, and so on.

IT as a "Business Innovation" Enabler

In today's market, many businesses understand how IT technologies enhance their services and provide innovation to their customers. Therefore, when a certain technology can serve as a business enabler that can help the organization to compete in the market or increase its customers' satisfaction, the adoption of that technology will be supported by the business [17].

For example, nowadays, cloud-based data centers are opening new opportunities for hosting service providers to generate more revenue for the business. To offer good cloud-based services, there must be a reliable, flexible, and high-performance data center infrastructure to deliver this service. Consequently, this engenders the initiative and will drive the business to build a high-performance, next-generation data center network. This network, by acting as a basis for cloud services, will be the enabler of the business's revenue-generation solution.

The Nature of the Business

Classifying the industry in which the business belongs or identifying the business's origins can aid in the understanding of some indirect requirements, even if these are not mentioned explicitly. For example, information security is almost always a must for a banking business whenever traffic crosses any external link. So by default, when planning a design for a business based in the banking industry, the design must support or offer security capabilities to gain acceptance from the business. In addition, industry-specific standards apply to IT infrastructure and services need to be considered. (For instance, healthcare organizations may consider complying with the IEC-80001-1 standard.[2])

2. http://www.iso.org/iso/catalogue_detail.htm?csnumber=44863

Business Priorities

Each business has a set of priorities that are typically based on strategies adopted for the achievement of goals. These business priorities can influence the planning and design of IT network infrastructure. Therefore, network designers must be aware of these business priorities to align them with the design priorities. This ensures the success of the network they are designing by delivering business value. For example, company X's highest priority is to provide a more collaborative and interactive business communication, followed by the provision of mobile access for the users. In this example, providing a collaborative and interactive communication must be satisfied first before providing or extending the communication over any mobility solution for the end users. In sum, it is important to align the design with the business priorities, which are key to achieving business success and transforming IT into a business enabler.

Functional Requirements

Functional requirements compose the foundation of any system design because they define system and technology functions. Specifically, functional requirements identify what these technologies or systems will deliver to the business from a technological point of view. For example, a Multiprotocol Label Switching (MPLS)-enabled service provider might explicitly specify a functional requirement in a statement like this: "The provider edge routers must send VoIP traffic over 10G fiber link while data traffic is to be sent over the OC-48 link." It is implied that this service provider network needs to have provider edge (PE) routers that support a mechanism capable of sending different types of traffic over different paths, such as MPLS Traffic Engineering (MPLS-TE). Therefore, the functional requirements are sometimes referred to as*behavioral requirements* because they address what a system does.

Note

The design that does not address the business's functional requirements is considered a poor design; however, in real-world design, not all the functional requirements are provided to the designer directly. Sometimes they can be decided on indirectly, based on other factors. See the "Application Requirements" section later in this chapter for more details.

Technical Requirements

The technical requirements of a network can be understood as the technical aspects that a network infrastructure must provide in terms of security, availability, and integration. These requirements are often called *nonfunctional requirements*. Technical requirements vary, and they must be used to justify a technology selection. In addition, technical requirements are considered the most dynamic type of requirements compared to other requirements such business requirements because, based on technology changes, they change often. Technical requirements include the following:

- Heightened levels of network availability (for example, using First Hop Redundancy Protocol [FHRP])
- Support the integration with network tools and services (for example, NetFlow Collector, or authentication and authorization system "RADIUS servers")
- Cater for network infrastructure security techniques (for example, control plane protection mechanisms or infrastructure access control lists [iACLs])

Note

The technical requirements help network designers to specify the required technical specifications (features and protocols) and software version that supports these specifications and sometimes influence the hardware platform selection based on its technical characteristics.

Application Requirements

From a business point of view, user experience is one of the primary, if not the highest, priority that any IT and network design must satisfy. The term *end users* can be understood differently according to the type of business. The following are the most common categories of end users:

- **Customers:** Customer can be individuals, such as a customer of a bank, or they can be a collective, such as the customers of an MPLS service provider. From a business point of view, customer satisfaction can directly impact the business's reputation and revenue.
- **Internal users:** In this category, the targeted users are internal users. Productivity of these users can translate to business performance efficiency, which has a direct relation to business success and revenue.
- **Business partners:** Partners represent those entities or organizations that work together to achieve certain goals. Therefore, efficient interaction between partners can enhance their business success in the service of strategic goal achievement.

Therefore, a network or a technology that cannot deliver the desired level of the users' expectation (also known as *quality of experience*) means a failure to achieve either one of the primary business goals or failure to satisfy a primary influencer of business success. Consequently, networks and IT technologies will be seen by the business as a cost center rather than as a business enabler.

On this basis, networks design must take into account how to deliver the desired level of experience. In fact, what influences users' experience is what they see and use. In other words, from a user's perspective, the quality of experience with regard to applications and services used by different types of users is a key deterministic factor.

Deploying network applications or services without considering the characteristics and network requirements of these applications will probably lead to a failure in meeting business goals. For example, a company providing modern financial services wants to distinguish itself from other competitors by enabling video communication with its customer service call center agents. If the network team did not properly consider video communication requirements as a network application, the application will probably fail to deliver the desired experience for the end users (customers in this example). Consequently, this will lead to a failure in achieving the company's primary business goal. In other words, if any given application is not delivered with the desired quality of experience to the end users, the network can simply be considered as not doing its job.

Furthermore, in some situations, application requirements can drive functional requirements. For instance, if a service provider has a service level agreement (SLA) with its clients to deliver Voice over IP (VoIP) traffic with less than 150 ms of one-way delay and less than 1 percent of packet loss, here VoIP requirements act as application requirements, which will drive the functional requirements of the PE devices to use a technology that can deliver the SLA. This technology may include, for example, a Class-Based Tunnel Selection (CBTS) MPLS-TE protected with fast reroute (FRR) to send VoIP traffic over high-speed links and provide network path protection in case of a failure within 50 ms. In addition, network designers should also consider the answers to the following fundamental questions when evaluating application requirements:

- How much network traffic does the application require?
- What is the level of criticality of this application and the service level requirement to the business?
- Does the application have any separation requirements to meet industry regulations and corporate security policies?
- What are the characteristics of the application?
- How long does the application need after losing connectivity to reset its state or session?

Design Constraints

Constraints are factors or decisions already in place and cannot be changed and often lead to a direct or indirect impact on the overall design and its functional requirements. In reality, various design constraints affect network design. The following are the most common constraints that network designers must consider:

- **Cost:** Cost is one of the most common limiting factors when producing any design; however, for the purpose of the CCDE practical exam, cost should be considered as a design constraint only if it is mentioned in the scenario as a factor to be considered or a tiebreaker between two analogous solutions.
- **Time:** Time can also be a critical constraint when selecting a technology or architecture over another if there is a time frame to complete the project, for example.
- **Location:** Location is one of the tricky types of constraints because it can introduce limitations that indirectly affect the design. For instance, a remote site may be located in an area where no fiber infrastructure is available and the only available type of connectivity is over wireless. From a high-level architectural point of view, this might not be a serious issue. From a performance point of view, however, this might lead to a reduced link speed, which will affect some sensitive applications used by that site.
- **Infrastructure equipment:** A good example here is that of legacy network devices. If a business has no plan to replace these devices, this can introduce limitations to the design, especially if new features or protocols not supported by these legacy platforms are required to optimize the design.
- **Staff expertise:** Sometimes network designers might propose the best design with the latest technologies in the market, which can help reduce the business's total cost of ownership (TCO) (for example, in the case of virtualization in the data center and the consolidation of data and Fibre Channel [FC] storage over one infrastructure Fibre Channel over Ethernet [FCoE]). This can be an issue, however, if the staff of this company has no expertise in these technologies used to operate and maintain the network. In this case, you have two possible options (with some limitations applying to each, as well):
- **Train the staff on these new technologies:** This will be associated with a risk, because as a result of the staff's lack of experience, they may take a longer time to fix any issue that might occur, and at the same time, data center downtime can cost the business a large amount of money.
- **Hire staff with experience in these technologies:** Normally, people with this level of expertise are expensive. Consequently, the increased operational cost might not justify the reduced TCO.

Note

In some situations, if the proposed solution and technologies will save the business a significant amount of money, you can justify the cost of hiring new staff.

Crafting the Design Requirements

This section demonstrates how different types of requirements collectively can lead to the achievement of the desired network design, which ultimately will facilitate achieving business goals. This demonstration is based on an example that goes through the flow of information gathering (the top-down approach, as illustrated in Figure 1-5) to build up the requirements starting from the business goals (top) to the technical requirement level, such as the required features (bottom).

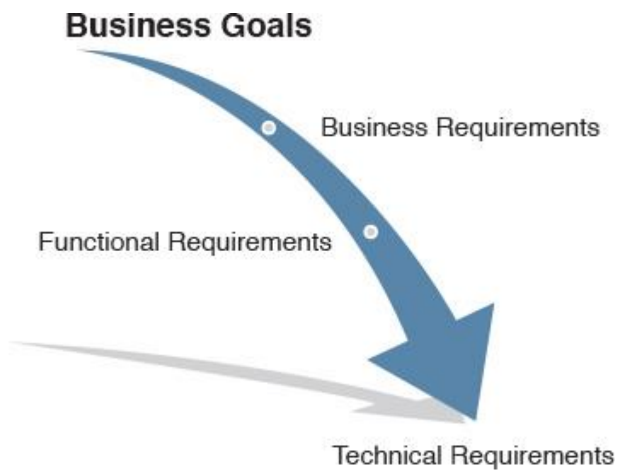

Figure 1-5 *Design Requirements Flow*

A national retail company is currently expanding their business to add several international sites within the next 12 months. However, with their current IT infrastructure, they face a high number of expenses in managing and maintaining their two current data and voice networks. In addition, the business wants to invest in technologies that offer enhancements to business activities and increase employee productivity.

The following is a typical classification of the requirements, some of which might be provided directly and some of which can be implied or indicated based on other requirements and goals:

- **Business goals:**
- Reduce operational cost
- Enhance employees' productivity
- Expand the business (adding more remote sites)
- **Business requirements:**
- Reduce the cost of maintaining multiple networks for voice and data
- Improve employee productivity by enhancing and integrating internal communications through video and mobile devices, without compromising the company's security policy
- Support the business expansion (the rollout of the new remote sites)
- **Functional requirements:**
- A unified infrastructure that supports voice, video, data, and wireless
- Ability to provide isolation between the traffic of guests and internal staff (for both wired and wireless) to comply with the standard security policy of the organization
- Capability to support introducing new remote sites to the network without any redesign
- **Application requirements:** In this particular example, we assume that no specific application requirements were given. In fact, they do not need to be provided because it is obvious that this organization is going to add VoIP and video as applications or services. The network must provide the required level of network efficiency, performance, and availability to meet VoIP/video requirements (real-time, delay, and jitter sensitive) to achieve one of the business goals.
- **Technical requirements:** To achieve the above network's functional and application requirements considering the ultimate business goals, the design must cater to the following:
- **High availability:** Increase high availability to support critical traffic such as voice (for example, redundant nodes, control plane tuning, FHRP)
- **Quality of service:** Improve users' quality of experience by introducing QoS to achieve high-quality voice and video services (for example, queuing, traffic shaping)
- **Security:** Optimize the network security to accommodate the new design, including voice and wireless security (for example, access control with 802.1X authentication)
- **Traffic isolation:** Provide traffic isolation to meet the information security requirements (for example, tunneling such as generic routing encapsulation [GRE] with MPLS/VRFs [virtual private network routers/forwarders])
- **Scalability:** Scalable network (WAN) design to support the projected business growth during the next 12 months (for example dynamic multipoint VPN [DMVPN] versus point-to-point GRE)

Table 1-2 summarizes the mapping between the technical requirements, business priorities, and technical implementation planning priorities.

Technical Requirements	Features/Tools to Enable the Technical Requirement	Business Priority Level, 1–5 (1 = highest, 5 = lowest)	Implementation Priorities
High availability	Modular and hierarchal network design. Network node and path redundancy. Tune control plane for fast convergence.	1	1
QoS	Traffic marking, classification, and prioritization to provide QoS.	2	3
Security	Add stateful firewalls with filtering to control communications between networks.	3	2
Traffic isolation	Use network virtualization techniques.	3	2
Scalability	WAN design must be scalable enough to support the business expansion such avoiding point-to-point mesh types of connectivity over the WAN.	1	1

Table 1-2 *Design Requirements Mapping*

Note that in Table 1-2 there are two levels of priorities:

- **Business priority level:** This level reflects the priority from business point of view.
- **Implementation priority level:** This level reflects the priorities from a technical implementation point of view. Although business priorities must always be met first, sometimes from a technical point of view there may be prerequisites to enable services or features across the network first to achieve a specific requirement. In the preceding example, from a technical design and implementation point of view, the network first needs to be designed and connected in the desired way (for example, hierarchical LAN, hub-and-spoke WAN). Following, the network designer can apply virtualization techniques to meet traffic separation requirements. At the same time, security appliances and functions must be integrated (holistic approach not siloed approach). Finally, QoS can be applied later to optimize traffic flows over both the physical and logical (virtualized) networks. In other words, the described sequence of the implementation plan in the preceding example was driven by both the business and technical priorities, considering that the business priorities must always be given the preference when technically possible.

Planning

There is an enduring adage: “If you do not have a plan, you are planning to fail.” This adage is accurate and applicable to network design. Many network designers focus on implementation after obtaining the requirements and putting them in a design format. They sometimes rely on the best practices of network design rather than focusing on planning: “What are the possible ways of getting from point A to point B?” This planning process can help the designer devise multiple approaches or paths (design options). At this point, the designer can ask the question: Why? Asking *why* is vital to making a business-driven decision for the solution or design that optimally aligns with the business’s short- or long-term strategy or objective. In fact, the best practices of network design are always recommended and should be followed whenever applicable and possible.

However, reliance on best practices is more common when designing a network from scratch (greenfield), which is not common with large enterprises and service provider networks.

In fact, IT network architectures and building constrictions and architectures are quite similar in the way they are approached and planned by designers or consultants.

For example, five years ago, an investment company built a shopping mall in one of the large cities in Europe, which was architected and engineered based on the requirements at that time. Recently, the stakeholders requested the design to be reviewed and optimized to overcome the increased number of people visiting this shopping mall (tourist and locals), because this increase was not properly projected and planned for during the original design five years ago.

Typically, the architects and engineers will then evaluate the situation, identify current issues, and understand the stakeholders' goals. In other words, they gather business requirements and identify the issues. Next, they work on optimizing the existing building (this may entail adding more parking space, expanding some areas, and so forth) rather than destroying the current building and rebuilding it from scratch. However, this time they need to have proper planning to provide a design that fits current and future needs.

Similarly, with IT network infrastructure design, there are always new technologies or broken designs that were not planned well to scale or adapt to business and technology changes. Therefore, network designers must analyze business issues, requirements, and the current design to plan and develop a solution that can optimize the overall existing architecture. This optimization might involve the redesign of some parts of the network (for example, WAN), or it might involve adding a new data center to optimize BC plans. To select the right design option and technologies, network designers need to have a *planning approach* to connect the dots at this stage and make a design decision based on the information gathering and analysis stage. Ultimately, the planning approach leads to the linkage of design options and technologies to the gathered requirements and goals to ensure that the design will bring value and become a business enabler rather than a cost center to the business. Typical tools network designers use at this stage to facilitate and simplify the selection process are the decision tree and the decision matrix.

Decision Tree

The decision tree is a helpful tool that a network designer can use when needing to compare multiple design options, or perhaps protocols, based on specific criteria. For example, a designer might need to decide which routing protocol to use based on a certain topology or scenario, as illustrated in Figure 1-6.

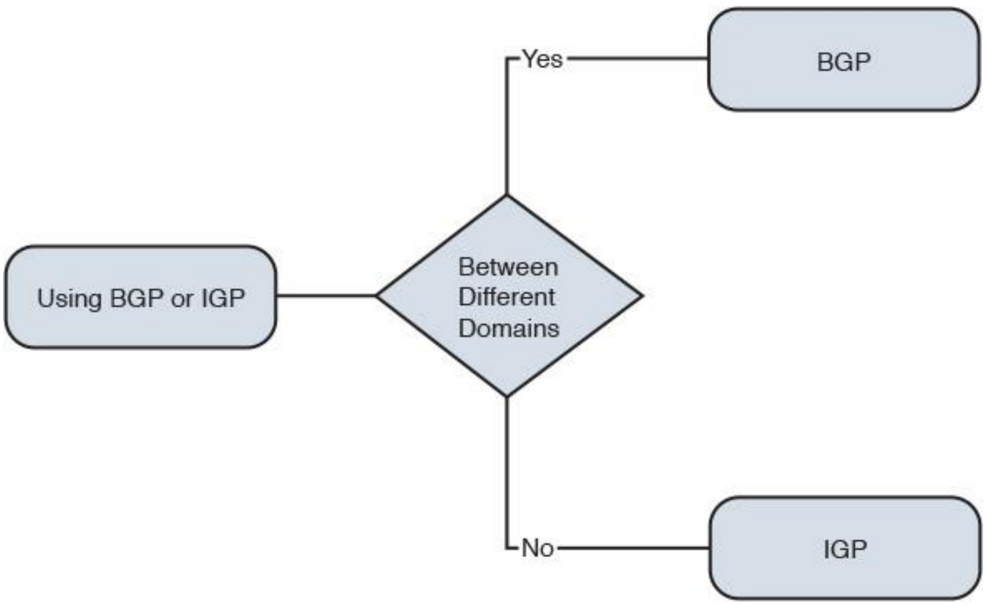

Figure 1-6 *Decision Tree*

Decision Matrix

Decision matrixes serve the same purpose as decision trees; however, with the matrix, network designers can add more dimensions to the decision-making process. In Table 1-3, there are two dimensions a network designer can use to select the most suitable design option. In these two dimensions, both business requirements and priorities can be taken into account to reach the final decision, which is based on a multidimensional approach.

Business Requirements	Priority	Design Option 1	Design Option 2
Cost savings	4	Moderate	Low
Scalable	1	High	High
Secure	2	Low	High

Table 1-3 *Decision Matrix*

When using the decision matrix as a tool in the preceding example, design option 2 is more suitable based on the business requirements and priorities. The decision matrix is not solely reliant on the business requirements to drive the design decision; however, priorities from the business point of view were considered as an additional dimension in the decision-making process, which makes it more relevant and focused.

Planning Approaches

To develop a successful network design, a proper planning approach is required to build a coherent strategy for the overall design. Network designers can follow two common planning approaches to develop business-driven network designs and facilitate the design decisions:

- **Strategic planning approach:** Typically targets planning to long-term goals and strategies. For example, a service provider needs to migrate its core from legacy (ATM) to be based on MPLS instead. The design decision in this approach has to cater to long-term goals and plans.

- **Tactical planning approach:** Typically targets planning to overcome an issue or to achieve a short-term goal. For instance, an enterprise might need to provide remote access to some partners temporally (for example, over the Internet) until dedicated extranet connectivity is provisioned. Design decisions in this approach generally need to provide what is required within a limited period of time and are not necessarily required to consider long-term goals.

Strategic Balance

Within any organization, there are typically multiple business units and departments. Each has its own strategy, some of which are financially driven, whereas others are more innovation driven. For example, an IT department is more of an in-house service provider concerned with ensuring service delivery is possible and optimal, whereas the procurement department is cost driven and always prefers the cheaper options. The marketing department, in contrast, is almost always innovation driven and wants the latest technology. Consequently, a good understanding of the overall business strategy and goals can lead to compromise between the different aims of the different departments. In other words, the idea is that each business unit or entity within an organization must have its requirements met at a certain level, so that all can collectively serve the overall business goals and strategies. For instance, let's consider a case study of a retail business wanting to expand its geographic presence by adding more retail shops across the globe with low capital expenditure (capex). Based on this goal, the main point is to increase the number of remote sites with minimal cost (expansion and cost):

- **IT point of view:**

- The point of sales application used does not support offline selling or local data saving. Therefore, it requires connectivity to the data center to operate.
- The required traffic volume from each remote site is small and non-real time.
- Many sites are to be added within a short period of time.
- **Optimum solution:** IT suggested that the most scalable and reliable option is to use a MPLS VPN as a WAN.
- **Marketing point of view:** If any site cannot process purchased items due to a network outage, this will impact the business's reputation in the market.
- **Optimum solution:** High-speed, redundant links should be used.
- **Finance point of view:** Cost savings.
- **Optimum solution:** One cheap link, such as an Internet link, to meet basic connectivity requirements.

Based on the preceding list, it is obvious that the consideration for WAN redundancy is required for the new remote sites; however, cost is a constraint that must be considered as well.

When applying the strategic balance (alignment) concept, each departmental strategy can be incorporated to collectively achieve the overall optimum business goal by using the suboptimum approach from each department's perspective.

In this particular example, you can use two Internet links combined with a VPN overlay solution to achieve the business goal through a cost-effective solution that offers link redundancy to increase the availability level of the remote sites, meeting application bandwidth requirements while at the same time maintaining the brand reputation in the market at the desired level.

Network Design Principles

Network design can be defined as the philosophy that drives how various components, protocols, and technologies should be integrated and deployed based on certain approaches and principles to construct a cohesive network infrastructure environment that can facilitate the achievement of tactical or strategic business goals. Previous sections in this chapter described the end-to-end process of a network design (PPDIOO) and the approaches that you can use to tackle different network designs (top down and bottom up).

This section, however, focuses on the design principles that network designers must consider when designing a network and translating business, functional, or application requirements into technological requirements. It is important to understand that these principles are not independent of each other. Therefore, to generate an effective design and achieve its intended goals, network designers should consider how each of these principles can integrate into the overall architecture; however, not all of these principles may be applicable or required by every design. By following the top-down approach, network designers can evaluate and decide what to consider and focus on.

Reliability and Resiliency

A reliable network delivers nearly every packet accepted by the network to the right destination within a reasonable amount of time [2]. In today's modern converged networks, maintaining a highly reliable network is extremely important because modern businesses rely heavily on IT services to achieve their business goals. For these businesses (especially service providers and modern financial institutions), it can be even more critical that their network is considered as a revenue-generating entity. For example, a five-minute network outage that might affect *x* number of customers can cost the business hundreds of thousands of dollars. Therefore, having a highly reliable and available network architecture that can survive during any network component failure without any operator intervention (also known as *resiliency*) is a key design principle. It is considered a top priority of most of today's modern businesses. For the network to achieve the desired level of resiliency, a redundant component must exist to take over the role of the failed component following a failure event.

Note

In spite of the fact that network availability is considered a "top priority of most of today's modern businesses," not every network needs high availability, and not every part of a network requires the same level of availability considerations. For more information about high-availability considerations, see Chapter 9, "Network High-Availability Design."

Modularity

Modular design is commonly used in software development, where an application can be built from multiple blocks of codes that collectively integrate to form the desired application. These "building blocks" of modular structure enhance and simplify the overall application architecture. For example, if an issue exists in one block or module of that software, it can easily be isolated from the other modules and fixed separately without impacting other parts. Furthermore, from the perspective of ongoing enhancements, it is easier to add additional modules or blocks to this structure if new features are required. This makes the overall application architecture more structured and manageable.

Similarly, modularity is one of the fundamental principles of a structured network. In a structured network, the network architecture can be divided into multiple functional modules, with each module serving a specific role in the network and represented by an individual physical network. The individual physical network is also known as the places in the network (PIN), such as the enterprise campus, WAN, or the data center.

Consequently, these functional modules are easy to replicate, redesign, and expand. Studies show that when building an IT network, about 20 percent of the budget goes to acquiring the hardware, and 80 percent goes to operational costs [9]. For instance, if a network is designed in a way (for example, flat) that cannot isolate security breaches, system upgrades, or failures in certain parts of the network, it will not be "responsive enough" to adapt to future business requirements such as scalability and fast network convergence.

Whereas with the modular design approach, if any given module faces an issue such as a security breach or the addition or removal of modules, there should be no need to redesign the network or introduce any effect to the other modules. Furthermore, breaking complex parts in the network based on a modular approach into manageable blocks will optimize the overall network manageability. In other words, from a network design point of view, modularity can promote design simplicity, flexibility, fault isolation, and scalability, as shown in Figure 1-7 [10]. At the same time, modularity reduces operational costs and complexities.

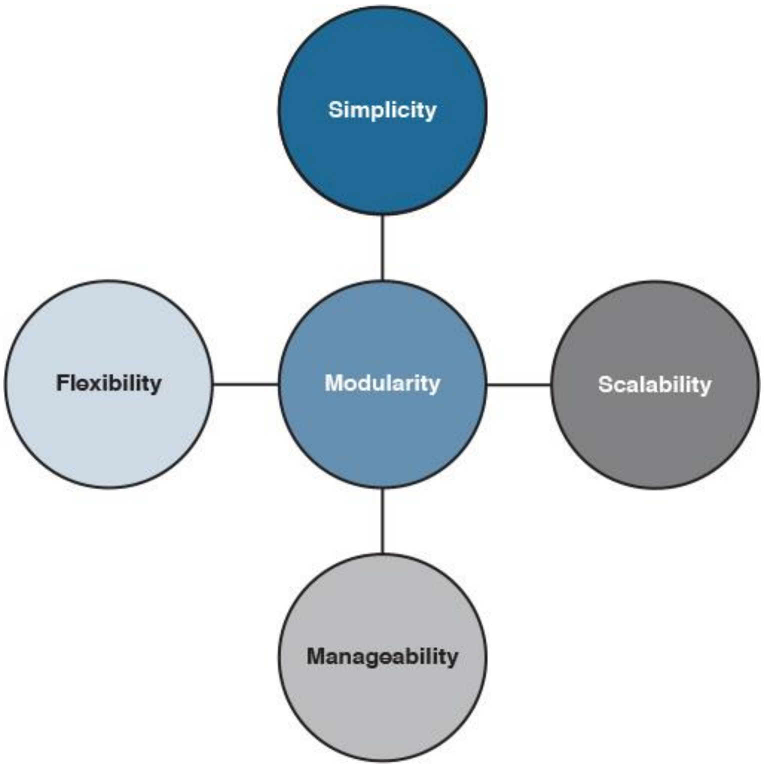

Figure 1-7 *Modularity*

Reliable and Manageable Scalability

Designing for scalability is one of the most important aspects that a network designer should consider. In fact, what can be considered as a successful scalable network is not only measured by the network size factor but also by how stable and reliable the network is and how easily this network can be managed and troubleshot. For example, a network may be designed in a way that can expand to thousands of routers, but at the same time, a failure in one link or device can introduce a high degree of processing and CPU utilization across the network, which may lead to instability. This design cannot be considered a successful scalable design, even though it can grow to a large number of routers. Similarly, a network may be designed to grow to a large scale while at the same time being difficult to manage and troubleshoot, with any issue potentially taking a long time to be fixed. This network cannot be considered a successful scalable design either.

Therefore, a successful scalable design must offer a manageable and reliable network at the same time. In other words, the added complexity of the configurations and troubleshooting with the increased size to an unmanageable scale will outweigh any advantage of the ability to expand by the size factor only, as shown in Figure 1-8 [8].

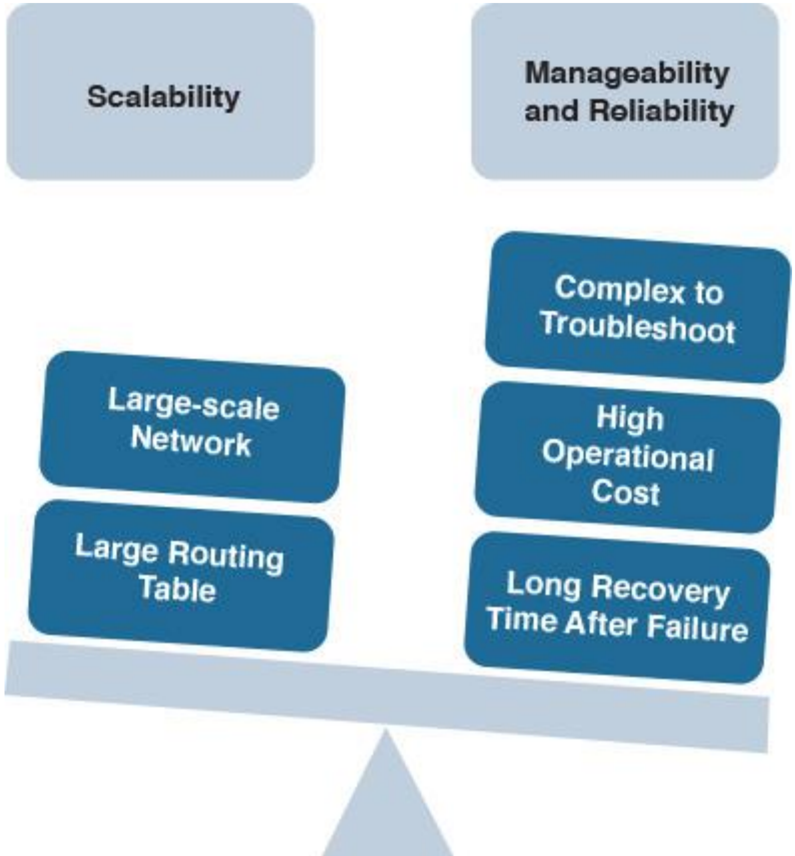

Figure 1-8 *Scalability Versus Manageability and Reliability*

Fault Isolation and Simplicity

Fault isolation boundaries are usually designed to provide fail-stop behavior for the desired fault model. The term *fail-stop* implies that incorrect behavior does not propagate across the fault isolation boundary [3]. In other words, network design must take into account how to prevent a failure in one domain from being signaled or propagated across all other domains in the network. This is necessary to avoid any unnecessary processing and delay across the entire network due to a link or device failure in one domain of the network. For example, in routing design, you can use summarization and logical flooding domains to mitigate this issue, where it is always advised that complicated networks (either physically or at the control plane layer) be logically contained, each in its own fault domain [10]. Consequently, you will have a more stable, reliable, and faster converging network.

Furthermore, this concept introduces simplicity to the overall architecture and reduces operational complexities. In general, fault isolation prevents network fault information from being propagated across multiple network areas or domains, which facilitates achieving more stable network design and optimizing network recovery time after any network failure event. For instance, with regard to the design in Figure 1-9, if any failure occurs on any network node or link, its impact will be propagated to all devices across the campus network. A high amount of unnecessary processing can occur after this failure.

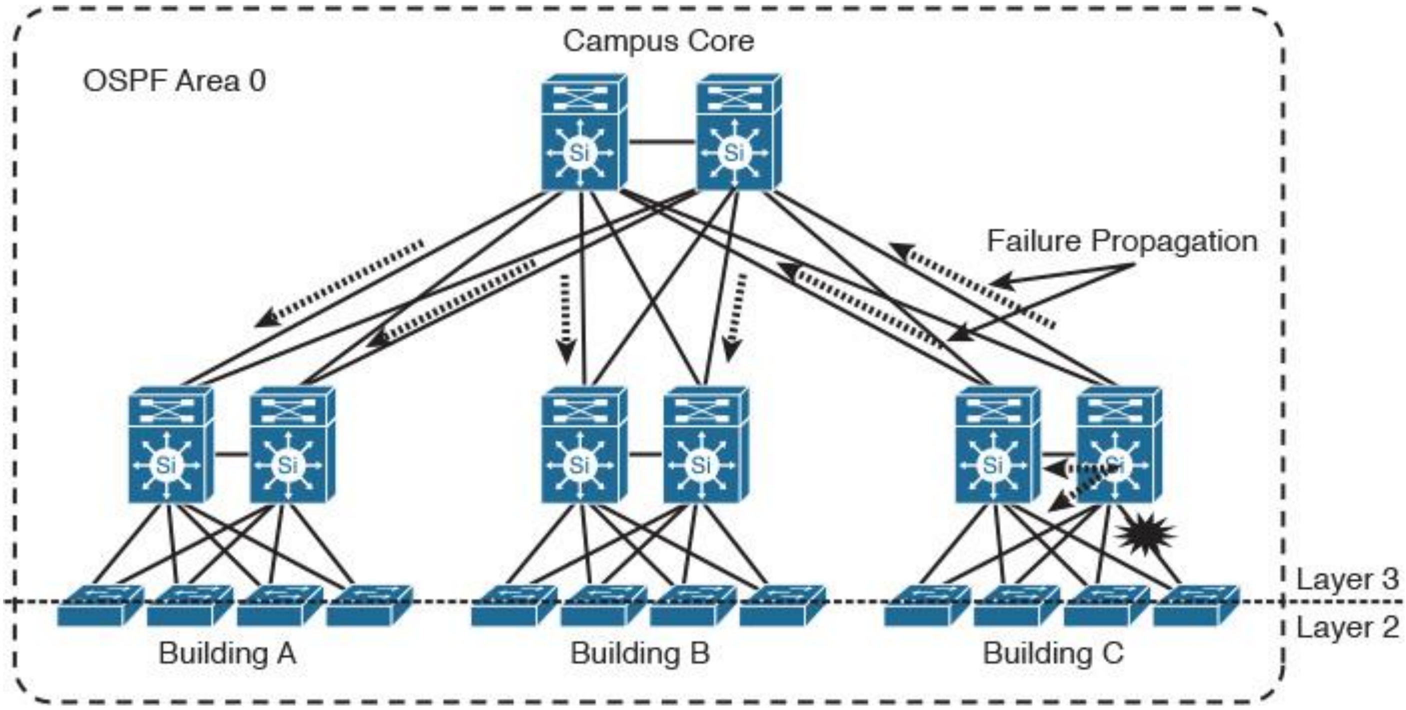

Figure 1-9 *Single-Fault Domain Design*

In contrast, with regard to the design in Figure 1-10, if there is any failure at any network node, there will be limited impact (typically contained within one fault domain). This limited impact is facilitated by the separation of the "fault domains" principle, which normally can be achieved by logical separation (for example, hiding reachability and topology information of Open Shortest Path First [OSPF] between the flooding domains or blocks). Consequently, the design will be able to offer a higher level of scalability, stability, and manageability.

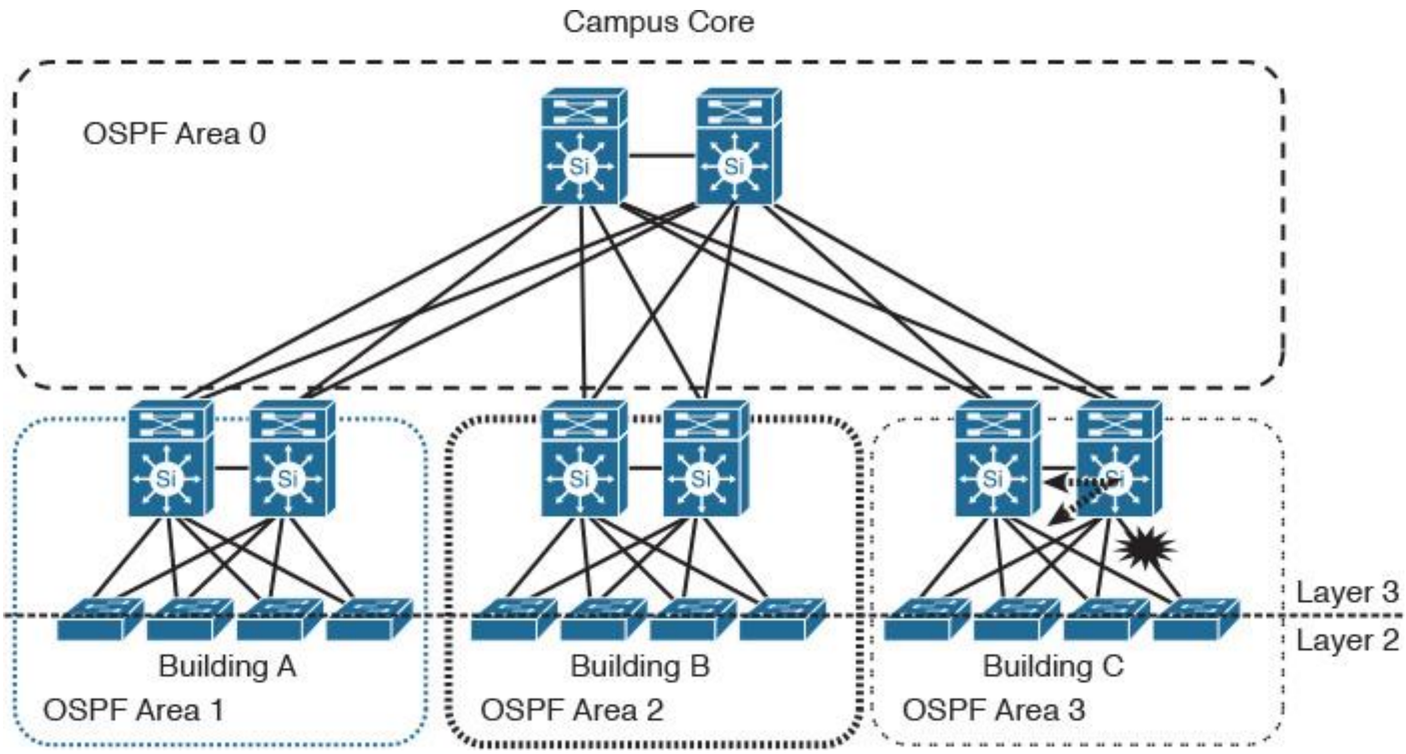

Figure 1-10 *Multi-Fault Domain Design*

Hierarchy

Adding hierarchy to the overall network design and within each domain or block can significantly simplify the design and enhance its flexibility to introduce more efficient places to isolate fault domains, because the network designer will have more chokepoints to aggregate traffic, topology information, and reachability information, which will facilitate the logical (functional) or physical isolation. For instance, Figure 1-11 shows a typical service provider network constructed of two tiers (core and POPs). Each POP is built from multiple tiers as well. This hierarchal structure provides the ability to control and minimize failure propagation between POPs over the core and between the different layers within each POP. Therefore, this design offers a level of stability (fault isolation) and control (chokepoints between the different tiers) to this service provider that makes it more responsive to any future growth requirements (scalable) without adding complexity to the overall design.

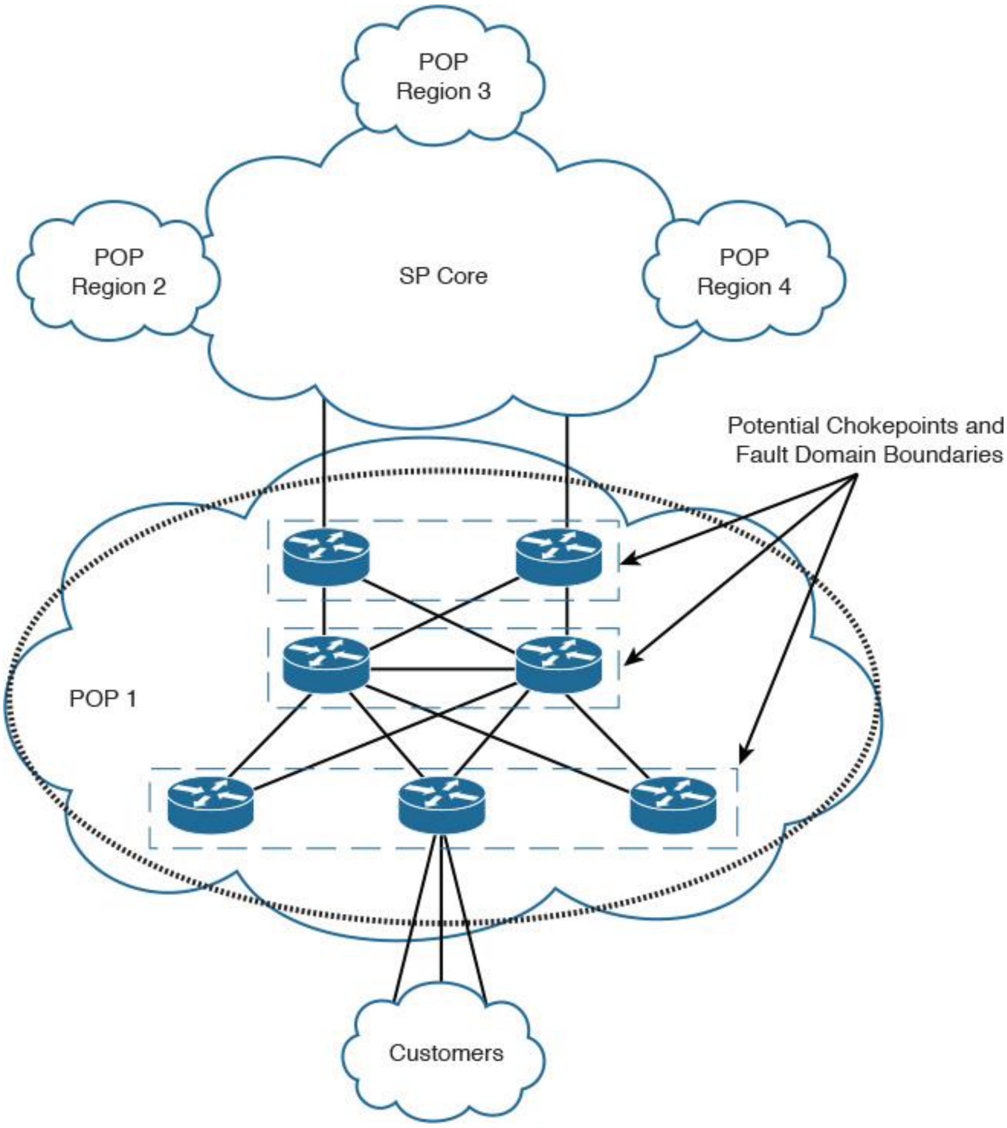

Figure 1-11 *Hierarchal Architecture*

Responsiveness

The term *responsiveness* refers to the set of design characteristics where a flexible architecture needs to respond to the changing business, application, and functional requirements. For example, the design should have the capability to support the rollout of secure mobility for internal staff and guests as well. In this case, a flexible architecture is required to respond to this new requirement and support rapid deployment and integration with different systems without any major change to the overall design. A modular and resilient design combined with the right forwarding and control plane architectures can collectively achieve this goal from the network point of view. When the overall architecture offers a high degree of flexibility, this will significantly facilitate the adoption and integration of the new business and technology requirements in the future, along with more simplified implementations to enable across the network. As a result, the network will be seen as a true service delivery entity from the business point of view.

Holistic Design Approach

A holistic design is a common design principle that considers a system being designed as an interconnected "whole" entity while at the same time potentially being part of something larger [4]. In other words, the holistic design approach emphasizes the big picture of any system, such as an enterprise network. Throughout this book, there are many design options and considerations of different types of networks, whether an enterprise network or service provider network, where each can fit a different set of requirements or solve certain issues. It is important to always consider the big picture using the holistic design concept before applying any of the designs discussed in this book in any given part of the network. Otherwise, you will be approaching the design using a siloed approach that will deal with the network as "communication islands." The result will possibly be a suboptimal architecture with lower performance, higher costs, and less flexibility. Most important, communications between these islands will probably be inefficient to a large extent because of the limited vision of this approach (not to mention the added complexity to manage networks that were not engineered as a single architecture).

To illustrate the importance of the holistic approach, consider the network in Figure 1-12. This network belongs to a manufacturing company that has a requirement to provide traffic separation between the internal staff and contractor staff from the remote sites across the WAN. A network designer suggested using two VRFs per site with a GRE tunnel per VRF across the WAN network. At the time of the design, there were only five remote sites. After six months, the number of remote sites doubled, and the company's IT staff started facing many complexities and traffic engineering issues. In addition, there was a requirement to extend this traffic separation to the data center services.

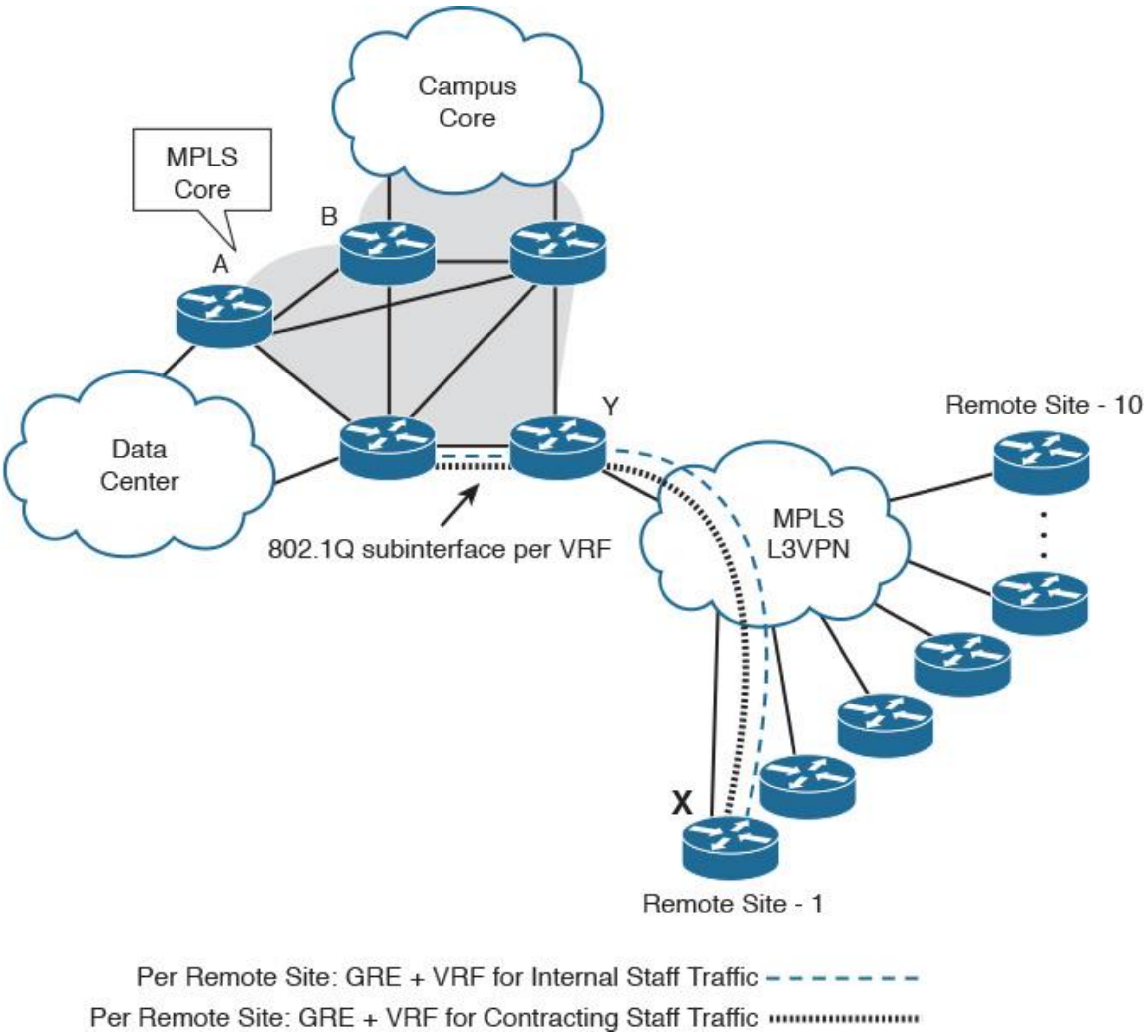

Figure 1-12 *Holistic Approach*

Technically, this is a valid design, and it does provide traffic isolation as per the business requirements; however, this design did not consider the big picture using a holistic design approach. Instead, it was designed based on communication islands (siloed approach), which probably will lead to a nonscalable and inflexible design.

In this particular example, the holistic approach could have helped the designer to consider the business growth requirements. It could have also helped the designer to consider how this traffic isolation can integrate and work as a whole interconnected system with the "MPLS-enabled enterprise core" for optimal and smooth end-to-end communications and integration. Instead, the designer focused only on how to meet this requirement between point X and Y. However, with the holistic approach, the designer can look at the entire architecture to see how point X and Y can optimally communicate with point A and B as well. If the number of remote sites increases, the designer can determine how this may affect the manageability and scalability of the entire network architecture, rather than focusing on only one side or block of the network.

Physical Layout Considerations

In building construction, the foundation carries the load of the structure on top of it, in addition to any future anticipated loads, such as people and furniture. Everything goes on top of the foundation. In other words, a solid and reliable foundation can be considered the basis of everything that comes after. Similarly, in network architectures, the foundation of the network (which is the underlying physical infrastructure) is where the entire traffic load is handled. It can be a critical influencer with regard to the adoption of certain designs or goals. An example of this is the interconnection of the service provider's core nodes over a physical fiber infrastructure in a ring topology. On top of this physical ring, there is a full mesh of Multiprotocol Border Gateway Protocol (MP-BGP) peering sessions between the PEs at the service provider edges (POPs), as illustrated in Figure 1-13.

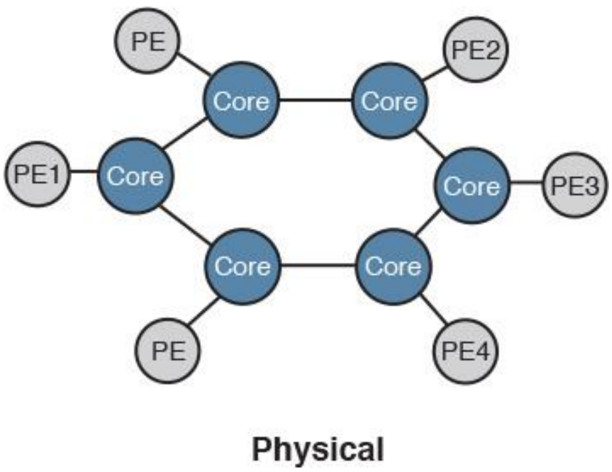

Figure 1-13 *Network Physical Layout View*

From the control plane and logical architectural point of view, these PEs are seen as directly interconnected, as shown inFigure 1-14.

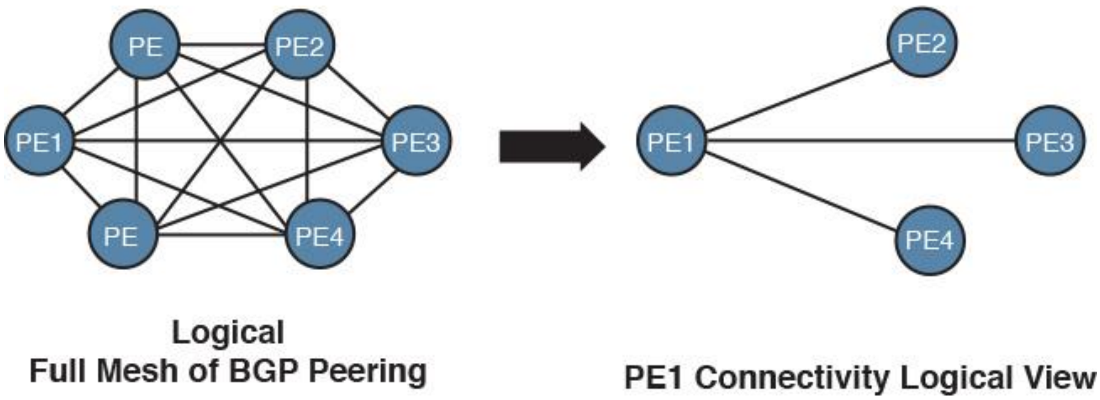

Figure 1-14 *Network Logical Layout View*

However, the actual traffic load and forwarding is handled by the physical core ring network, which means if any core router in the path has a congested link, it can affect all the POPs/PEs communicating over this path.

Consequently, it is important that network designers understand the foundation of the network to evaluate the load, traffic flow, and other aspects that sometimes cannot be seen from a protocol and the logical design point of view. In addition, understanding the underlying physical infrastructure can help network designers to steer traffic over the desired links based on different criteria such as expense of a given link, bandwidth requirements, and reliability of a given link to recover from a failure. (For example, protected fiber links can be more reliable than nonprotected links.) Throughout this book, the implications of using different Layer 2 and Layer 3 control protocols and overlay technologies are covered, considering the physical layout (the foundation of the network) when applicable. In other words, the logical layout of a network should never be designed in isolation of the underlying physical topology.

Note

Sometimes it is impossible, and unreasonable, to have visibility of the physical topology, such as routing or tunneling over the Internet.

Table 1-4 provides a summarized comparison between the most common physical network layouts.

Topology Layout	Scalability	Cost	Manageability	Flexibility	Redundancy
Ring	Limited	Low	Moderate	Limited due to the high level of dependencies	Limited
Fully meshed	Limited (introduce limitations on higher layers [for example, control plane])	High	Complex	Limited	Very high
Partially meshed	Moderate	Moderate	Moderate	Moderate	Moderate
Multiplane core	High	High	Simple	High	Moderate
Leaf and spines	Can offer a high degree of scale-out capability	Moderate	Simple	High	High
Hierarchal	Can offer high scale up	Moderate	Simple	High	High
Hub and spoke	Limited	Moderate	Moderate (depends on the number of spokes)	Limited	Moderate

Table 1-4 *Physical Topologies Comparison*

Note

Redundancy in Table 1-4 refers to the nature of the topology to support redundancy. Network designers must decide on the level of redundancy and in which areas redundancy is to be placed, based on the design requirements and goals.

No Gold Plating

Gold plating in software engineering or project management (or time management in general) refers to continuing to work on a project or task well past the point where the extra effort is worth the value it adds (if any)[4]. This concept is 100 percent applicable to network design and can be considered one of the critical principles of network design. In fact, overengineering or adding more features and capabilities that go beyond the requirements (whether business, functional, or application) can be a double-edged sword. In other words, a network designer might think that adding extra features to the design will make the customer more satisfied. However, these extra features may lead to additional implementation time and cost (for example, higher end-product prices or more licenses), whereas the design without these features still meets the requirements. Therefore, network designers must produce a design that focuses on how to meet the requirements and become a business enabler rather than add unnecessary features, which can lead to a negative impact.

Note

Requirements (business, functional, or application) define what is necessary (and what is not). As discussed earlier, the requirements can take different forms, such as fixing a current limitation or providing a design that will align with the business's future vision.

Summary

To achieve a network design that delivers value to the business and aligns with its goals and directions, network designers must follow a structured approach. This approach must start from the top level, focusing on business needs, drivers, and directions, to generate a business-driven design. In addition, with the top-down approach, network designers and architects can always produce business-driven network architecture that facilitates, at later stages, the selection of the desired hardware platforms and technology features and protocols to deploy the intended design. This makes the network design more responsive to any new business or technology requirements. Furthermore, considering the different design principles discussed in this chapter and considering the different requirements (business, functional, and mission-critical applications) can help network designers to plan and make the right design choices that ultimately will make the network be seen as a “business enabler.”

Part II: Next Generation - Converged Enterprise Network Architectures

Many businesses today rely on technology to achieve their strategic goals. Therefore, today's enterprise networks are evolving to be more like a "service delivery" mechanism for different business services and applications, in particular real-time enterprise applications and services, over one converged infrastructure. Traditionally, these services used to be provisioned over a separate physical infrastructure, such as voice and video. In contrast, a converged network logically consolidates these physical networks into one unified physical network with multiple logical networks and allows for optimal connectivity between these various logical networks.

The typical question network designers may ask, "Is there any difference in how converged networks are designed today?"

In fact, this question is one of the key design questions. However, let's first review the applications that today's converged enterprise networks are carrying and the requirements and level of criticality of these applications.

	Collaboration Applications				Data Center Applications		Multicast-Based Apps	
Network Requirement	Today's Apps	Voice Bearer	Video Bearer	WebEx	Storage	VM, Live Migration	Live Video	Market Feeds
Latency	< 500 ms	< 250 ms	< 250 ms	< 250 ms	5 ms (S) 100 ms (A)	< 200 ms	< 15 ms	< 20 ms
Jitter	< 200 ms	< 50 ms	< 50 ms	< 100 ms	< 100 ms	< 100 ms	< 100 ms	Various < 10 ms
Bandwidth per flow or instance	Various	40 Kbps (G729A)	4 Mbs (1080p)	300 Kbps (15 fps)	Various (4/6/8 Gbps)	Various 25000 pps	Various (10 Gbps plus)	Various > 80 Mbps per feed
Convergence	< 10 sec	< 1 sec	< 1 sec	< 10 sec	< 500 ms (A) < 100ms (S)	< 2 sec	< 2 sec	< 2 sec

Enterprise Application Requirements

It is obvious from the preceding table[6] that most of these applications are critical to the daily business activities. However, these advanced applications and technologies bring aggressive requirements from network performance and resiliency perspectives. Traditionally, the network used to be seen by the business as a cost center. Today's converged network has been transformed from being seen as a cost center to being seen as a business enabler. As a result, the requirements to design a converged enterprise network are changing, as elucidated upon in the following examples:

- Converged critical applications on one shared network (voice, video, and data).
- High availability and fast converging. The downtime cost of a converged network is very high compared to a traditional network.
- Low latency/low jitter for real-time communication services.
- Scalable (hundreds of nodes, global scale).
- System virtualization.
- Mobility.
- Integrated infrastructure security.

- Maintenance with minimal impact (ideally zero).

The requirements of a today's converged enterprise networks are driving the transformation of network design considerations and approaches.

Note

For the purpose of the CCDE exam, you must consider only the business drivers and requirements that are provided to you in each scenario. For example, do not assume cost saving, scalability, or high availability are always required if not mentioned as a requirement, design goal, or possible solution to optimize the design to overcome an existing issue or limitation.

This section discusses the various design principles and aspects of the next-generation converged enterprise network architectures. These network architectures use a modular design approach to divide the network into multiple functional areas that are referred to as *modules* or *blacks* (interchangeably). The modularity that is built in to the architecture allows flexibility and scalability in network design and facilitates the future deployment and operation of the network.

There can be different types of functional modules introduced as part of this architecture based on the business needs, size, and industry. This book discusses the most common and vital modules that can be found in almost every large-scale converged enterprise network. These modules are as follows:

- Enterprise campus
- Enterprise edge (WAN and Internet Blocks)
- Enterprise data center (The data center is discussed in Chapter 8, "Data Center Network Design.")

Chapter 2, "Enterprise Layer 2 and Layer 3 Design," starts by discussing Layer 2 and Layer 3 network design considerations. It specifically discusses the implications with regard to the different infrastructure topologies that collectively construct the overall Layer 2 and Layer 3 converged enterprise network architecture. The subsequent chapters focus on the design considerations and options of each module of the enterprise network without compromising the holistic design approach. In other words, although each enterprise module is discussed separately, the best design option must always be selected based on both the top-down and holistic approaches. This avoids the designing of the enterprise network modules in isolation and ensures that the design is always driven by the business and application requirements.

Chapter 2. Enterprise Layer 2 and Layer 3 Design

In network design, it is common that a certain design goal can be achieved "technically" using different approaches. Although, from a technical deployment point of view this can be seen as an advantage, from networks' design perspective on the other hand, almost always one of the most challenging parts is, which design option should be selected? (To achieve a business driven design that takes into considerations technical and non-technical design requirements.) Practically, to achieve this, network designers must be aware of the different design options and protocols as well as the advantages and limitations of each. Therefore, this chapter will concentrate specifically on highlighting, analyzing, and comparing the various design options, principles, and considerations with regard to Layer 2 and Layer 3 control plane protocols from different design aspects, focusing on enterprise grade networks.

Enterprise Layer 2 LAN Design Considerations

To achieve a reliable and highly available Layer 2 network design, network designers need to have various technologies in place that protect the Layer 2 domain and facilities having redundant paths to ensure continuous connectivity in the event of a node or link failure. The following are the primary technologies that should be considered when building the design of Layer 2 networks:

- Layer 2 control protocols, such as Spanning Tree Protocol
- VLANs and trunking
- Link aggregation
- Switch fabric (discussed in Chapter 8, "Data Center Networks Design")

Spanning Tree Protocol

As a Layer 2 network control protocol, the Spanning Tree Protocol (STP) is considered the most proven and commonly used control protocol in classical Layer 2 switched network environments, which include multiple redundant Layer 2 links that can generate loops. The basic function of STP is to prevent Layer 2 bridge loops by blocking the redundant L2 interface to a level that can provide a loop-free topology. There are multiple flavors or versions of STP. The following are the most commonly deployed versions:

- **802.1D:** The traditional STP implementation
- **802.1w:** Rapid STP (RSTP) supports large-scale implementations with enhanced convergence time
- **802.1s:** Multiple STP (MST) permits very large-scale STP implementations

In addition, there are some features and enhancements to STP that can optimize the operation and design of STP behavior in a classical Layer 2 environment. The following are the primary STP features:

- **Loop Guard:** Prevents the alternate or root port from being elected unless bridge protocol data units (BPDUs) are present
- **Root Guard:** Prevents external or downstream switches from becoming the root
- **BPDU Guard:** Disables a PortFast-enabled port if a BPDU is received
- **BPDU Filter:** Prevents sending or receiving BPDUs on PortFast-enabled ports

Figure 2-1 briefly highlights the most appropriate place where these features should to be applied in a Layer 2 STP-based environment.

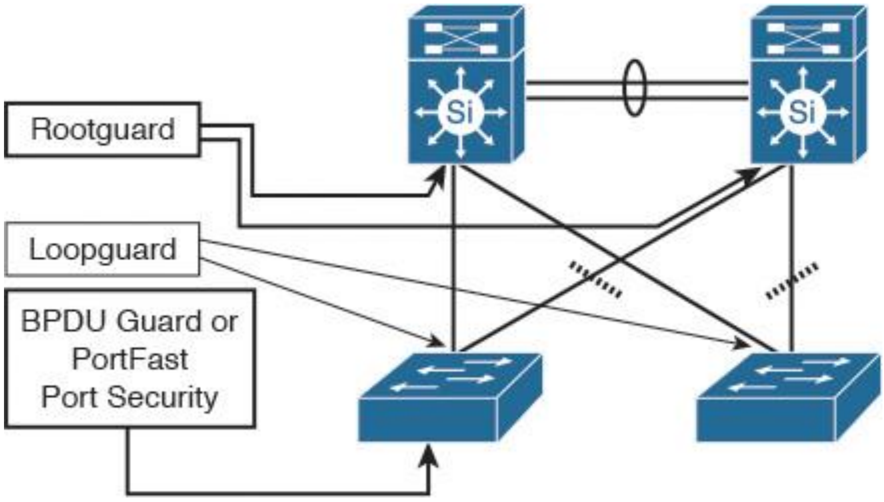

Figure 2-1 *STP Features*

Note

Cisco has developed enhanced versions of the STP. It has incorporated a number of these features into it using different versions of STP that provide faster convergence and increased scalability, such as Per-VLAN Spanning Tree Plus (PVST+) and Rapid PVST+.

VLANs and Trunking

A Layer 2 virtual local-area network (VLAN) is considered as a type of network virtualization technique that provides logical separation with broadcast domains and policy control implementation. In addition, VLANs offer a degree of fault isolation at Layer 2 that can contribute to the optimization of network performance, stability, and manageability.*Trunking*, however, refers to the protocols that enable the network to extend VLANs across Layer 2 uplinks between different nodes by providing the ability to carry multiple VLANs over a single physical link.

From a design best practices perspective, VLANs should *not* span multiple access switches; however, this is only a general recommendation. For example, some designs dictate that VLANs must span multiple access switches to meet certain application requirements. Consequently, understanding the different Layer 2 topologies and the impact of spanning VLANs across multiple switches is a key aspect for Layer 2 design. This aspect (applicability and its implications) is covered in more detail later in this chapter (in the section "Enterprise Layer 2 LAN Common Design Options").

Link Aggregation

The concept of link aggregation refers to the industry standard IEEE 802.3ad, in which multiple physical links can be grouped together to form a single logical link. This concept offers a cost-effective solution by increasing cumulative bandwidth without requiring any hardware upgrades. The IEEE 802.3ad Link Aggregation Control Protocol (LACP) offers several other benefits, including the following:

- An industry standard protocol that enables interoperability of multivendor network devices
- The optimization of network performance in a cost-effective manner by increasing link capacity without changing any physical connections or requiring hardware upgrades
- Eliminate single points of failure and enhance link-level reliability and resiliency

Although link aggregation is a simple and reasonable mechanism to increase bandwidth capacity between network nodes, each individual flow will be limited to the speed of the utilized member link by that flow, based on the load-balancing hashing algorithm used unless the flowlet concept is considered[1].

1. "Dynamic Load Balancing Without Packet Reordering," IETF Draft, chen-nvo3-load-banlancing, http://www.ietf.org

Note

In addition to LACP (the industry standard link aggregation control protocol), Cisco has developed a proprietary link aggregation protocol called Port Aggregation Protocol (PAgP). Both protocols have different operational modes, which the network designer must be aware.

There are two primary types of link aggregation connectivity models:

- **Single-chassis link aggregation:** The typical link aggregation type of connectivity that connects two network nodes in a point-to-point manner.
- **Multichassis link aggregation:** Also referred to as mLAG, this type of link aggregation connectivity is most commonly used when the upstream switches (typically two) are deployed in "switch clustering" mode. This connectivity model offers a higher level of link and path resiliency than the single-chassis link aggregation.

Figure 2-2 illustrates these two link aggregation connectivity models.

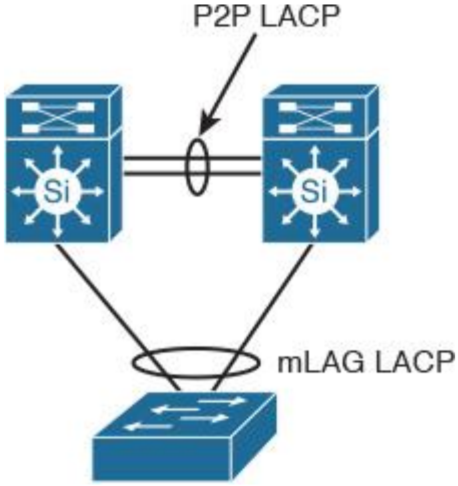

Figure 2-2 *Link Aggregation*

First Hop Redundancy Protocol and Spanning Tree

First-hop Layer 3 routing redundancy is designed to offer transparent failover capabilities at the first-hop Layer 3 IP gateways, where two or more Layer 3 devices work together in a group to represent one virtual Layer 3 gateway. Hot Standby Router Protocol (HSRP), Virtual Router Redundancy Protocol (VRRP), and Gateway Load Balancing Protocol (GLBP) are the primary and most commonly used protocols to provide a resilient default gateway service for endpoints and hosts.

Table 2-1 summarizes and compares the main capabilities and functions of these different FHRP protocols.

	HSRP	VRRP	GLBP
Standard	Cisco proprietary	IEEE	Cisco proprietary
IPv6 support	Yes (v2)	Yes (v3)	Yes
Authentication	Yes	Yes	Yes
Load sharing techniques	Multiple HSRP groups per interface	Multiple VRRP groups per interface	Native support (implicit)
Default hello timer	3	1	3
Virtual IP VIP	Different from interface IP	Can be different or same to the interface IP of the master router	Different from the interface IP
BFD support for subsecond convergence	Yes	Yes	No

Table 2-1 *FHRP Protocols Comparison*

One of the typical scenarios in classical hierarchal networks is when FHRP works in conjunction with STP to provide a redundant Layer 3 gateway services. However, some Layer 2 design models require special attention in terms of VLANs design (extending Layer 2 VLANs across access switches or not) and the placement of the demarcation point between Layer 2 and Layer 3. (For example, if the interswitch link between the distribution layer switches is configured as Layer 2 or Layer 3 link.) The aforementioned factors can affect the overall reliability and convergence time of the Layer 2 LAN design. The following design model (depicted in Figure 2-3) is considered one of the common design models that has a proven ability to provide the most resilient design when FHRP is applied to an STP-based Layer 2 network (such as VRRP or HSRP). This design model has the following characteristics:

- The interswitch link between the distribution switches is configured as Layer 3 link.
- No VLAN spanning across switches.
- The STP root bridge is aligned with active FHRP instance for each VLAN.
- Uplinks from access to distribution are both forwarding from STP point of view.

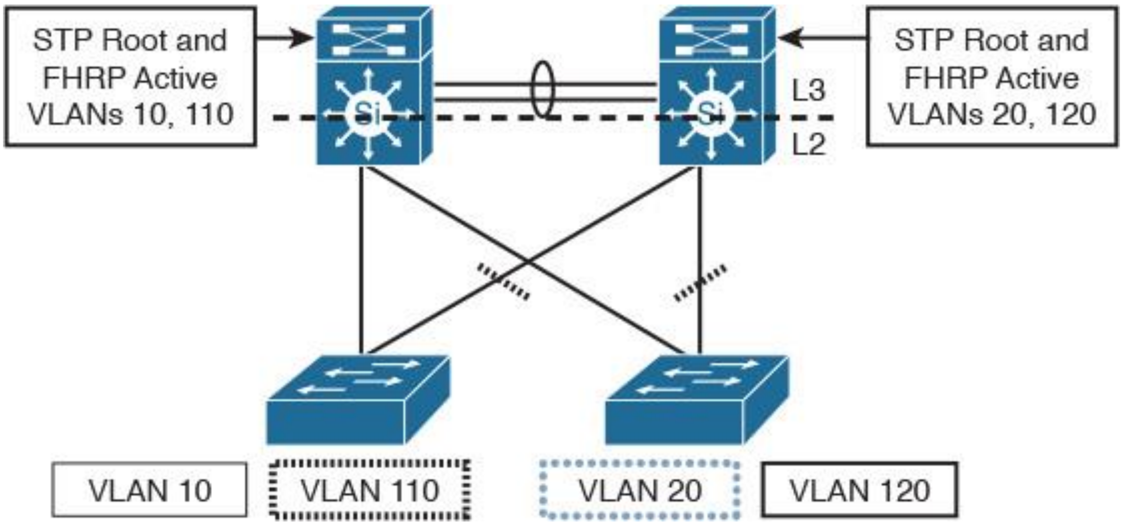

Figure 2-3 *FHRP and STP on Loop-Free U Topology*

Note

In the design illustrated in Figure 2-3, when GLBP is used as the FHRP, it is going to be less deterministic compared to HSRP or VRRP because the distribution of Address Resolution Protocol (ARP) responses is going to be random.

Enterprise Layer 2 LAN Common Design Options

Network designers have many design options for Layer 2 LANs. This section will help network designers by highlighting the primary and most common Layer 2 LAN design models used in traditional and today's LANs, along with the strengths and weaknesses of each design model.

Layer 2 Design Models: STP Based (Classical Model)

In classical Layer 2 STP-based LAN networks, the connectivity from the access to the distribution layer switches can be designed in various ways and combined with Layer 2 control protocols and features discussed earlier to achieve certain design functional requirements. In general, there is no single best design that someone can suggest that can fit every requirement, because each design is proposed to resolve a certain issue or requirement. However, by understanding the strengths and weaknesses of each topology and design model (illustrated in Figure 2-4), network designers may then always select the most suitable design model that meets the requirements from different aspects, such as network convergence time, reliability, and flexibility. This section highlights the most common classical Layer 2 design models of LAN environments with STP, which can be applied to enterprise Layer 2 LAN designs.

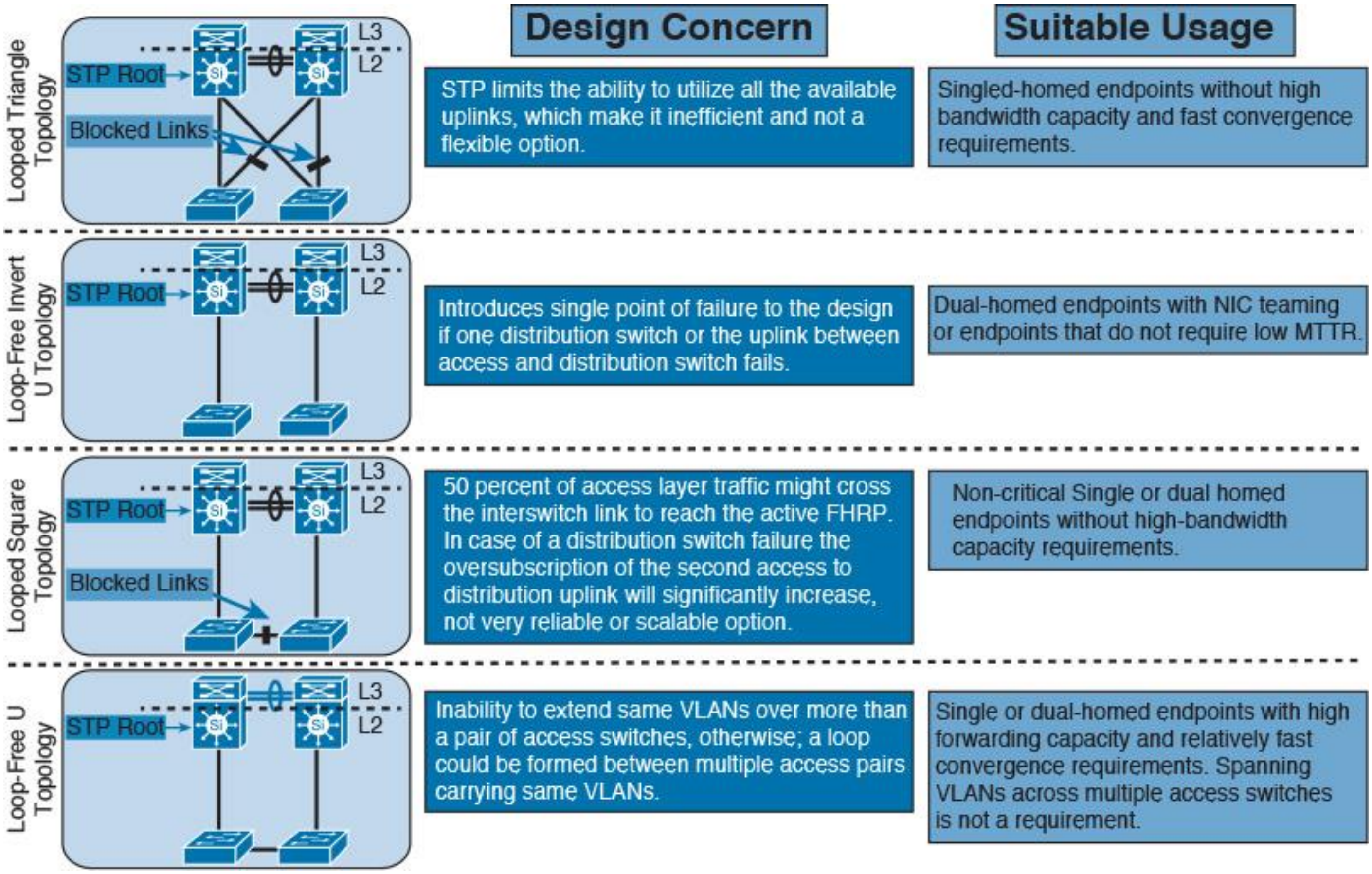

Figure 2-4 *Primary and Common Layer 2 (STP-Based) LAN Connectivity Models Comparison*

Note

All the Layer 2 design models in Figure 2-4 share common limitations: the reliance on STP to avoid loss of connectivity caused by Layer 2 loops and the dependency on Layer 3 FHRP timers, such as VRRP to converge. These dependences naturally lead to an increased convergence time when a node or link fails. Therefore, as a rule of thumb, tuning and aligning STP and FHRP timers is a recommended practice to overcome these limitations to some extent.

Figure 2-4 summarizes some of the design concerns and lists suggested usage of each of the depicted design models in this figure [5].

Layer 2 Design Model: Switch Clustering Based (Virtual Switch)

The concept of switch clustering significantly changed the Layer 2 design model between the access and distribution layer switches. With this design model, a pair of upstream distribution switches can appear as a one logical (virtual) switch from the access layer switch point of view. Consequently, this approach transformed the way access layer switches connect to the distribution layer switches, because there is no reliance on STP and FHRP anymore, which means the elimination of any convergence delays associated with STP and FHRP. In addition, from the uplinks and link aggregation perspective, one access switch can be connected (multihomed) to the two clustered distribution switches as one logical switch using one link aggregation bundle over multichassis link aggregation (mLAG), as illustrated in Figure 2-5.

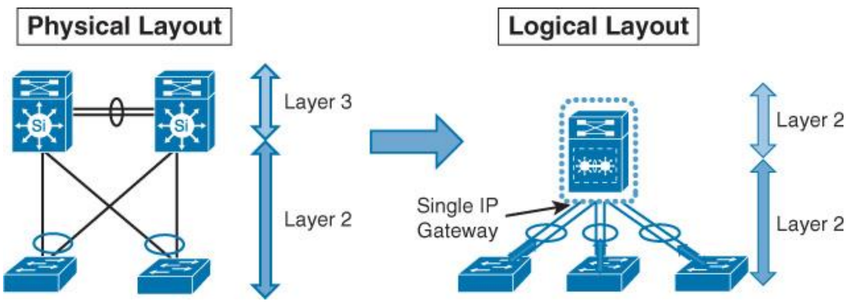

Figure 2-5 *Switch Clustering*

As Figure 2-5 shows, all uplinks will be in forwarding state across both distribution switches from a Layer 2 point of view. There will be one virtual IP gateway that should permit the forwarding across both switches from the forwarding plane perspective. It is obvious that this design model can enhance network resiliency and convergence time, and maximize bandwidth capacity, by utilizing all uplinks. In addition, this design model supports the extension of the Layer 2 VLAN across access switches safely, without any concern about forming any Layer 2 loop. This makes the design model simple, reliable, easy to manage, and more scalable as compared to the classical STP-based design model.

Layer 2 Design Model: Daisy-Chained Access Switches

Although this design model might be a viable option to overcome some limitations, network designers commonly use it as an interim solution. This design can introduce undesirable network behaviors. For instance, the design shown in Figure 2-6 can introduce the following issues during a link or node failure:

- Dual active HSRP
- Possibility of 50 percent loss of the returning traffic for devices that still use the distribution switch-1 as the active Layer 3 FHRP gateway

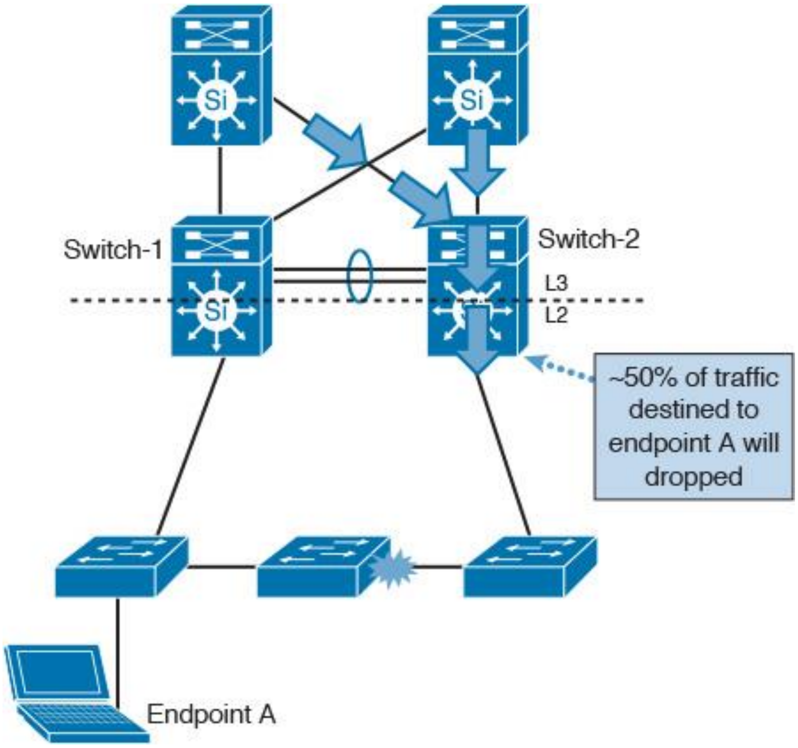

Figure 2-6 *Daisy-Chained Access Switches*

When suggesting an alternative solution to overcome a given design issue or limitation, it is important to make sure that the suggested design option will not introduce new challenges or issues during certain failure scenarios. Otherwise, the newly introduced issues will outweigh the benefits of the suggested solution.

Layer 2 LAN Design Recommendations

Table 2-2 summarizes the different Layer 2 LAN design considerations and the relevant design recommendations.

Design Considerations	Design Recommendations
Flexibility and scalability	Use switch clustering at the distribution layer when possible. Consider using link aggregation or mLAG between the different LAN layers. Avoid spanning L2 VLANs across multiple access switches.
Resiliency and Fast convergence	Design with triangles and not square topologies when possible. Always use switch clustering with mLAG when possible. Align the STP root with FHRP active when using the traditional L2 design model. Tune STP and FHRP timers. Use Unidirectional Link Detection (UDLD). Always interconnect the upstream distribution switch.
Control and security	Always enable STP, even with switch clustering. Use BBDU Guard, filtering, and Root Guard. Use an unused VLAN as the native VLAN. Use port-level security features such as Dynamic Host Configuration Protocol (DHCP) snooping, IP Source Guard, and ARP. security where applicable

Table 2-2 *Layer 2 LAN Design Recommendation*

Enterprise Layer 3 Routing Design Considerations

This section covers the various routing design considerations and optimization concepts that pertain to enterprise-grade routed networks.

IP Routing and Forwarding Concept Review

The main goal of routing protocols is to serve as a delivery mechanism to route packets to reach their intended destination. The end-to-end process of packets routing across the routed network is facilitated and driven by the concept of distributed databases. This concept is typically based on having a database of IP addresses (typically IPs of hosts and networks) on each Layer 3 node in the packet's path, along with the next-hop IP addresses of the Layer 3 nodes that can be used to reach each of these IP addresses. This database is known as the *Routing Information Database* (RIB). In contrast, the Forwarding Information Base (FIB), also known as the *forwarding table*, contains the destination addresses and the interfaces required to reach those destinations, as depicted in Figure 2-7. In general, routing protocols are classified as either link-state, path-vector, or distance-vector protocols. This classification is based on how the mechanism of the routing protocol constructs and updates its routing table, and how it computes and selects the desired path to reach the intended IP destination.[2]

2. IETF draft: (Routing Information Base Info Model "draft-nitinb-i2rs-rib-info-model-02")

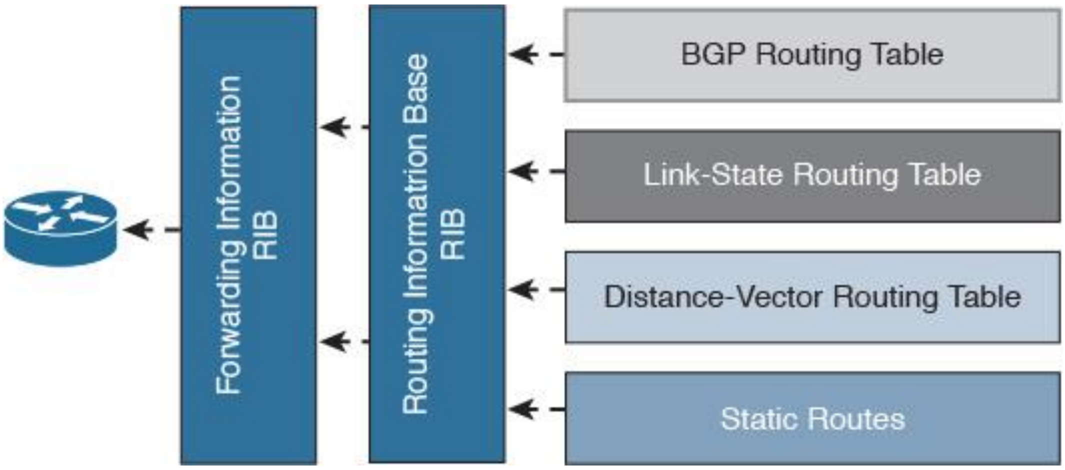

Figure 2-7 *RIB and FIB*

As illustrated in Figure 2-8, the typical basic forwarding decision in a router is based on three processes:

- Routing protocols
- Routing table
- Forwarding decision (switches packets)

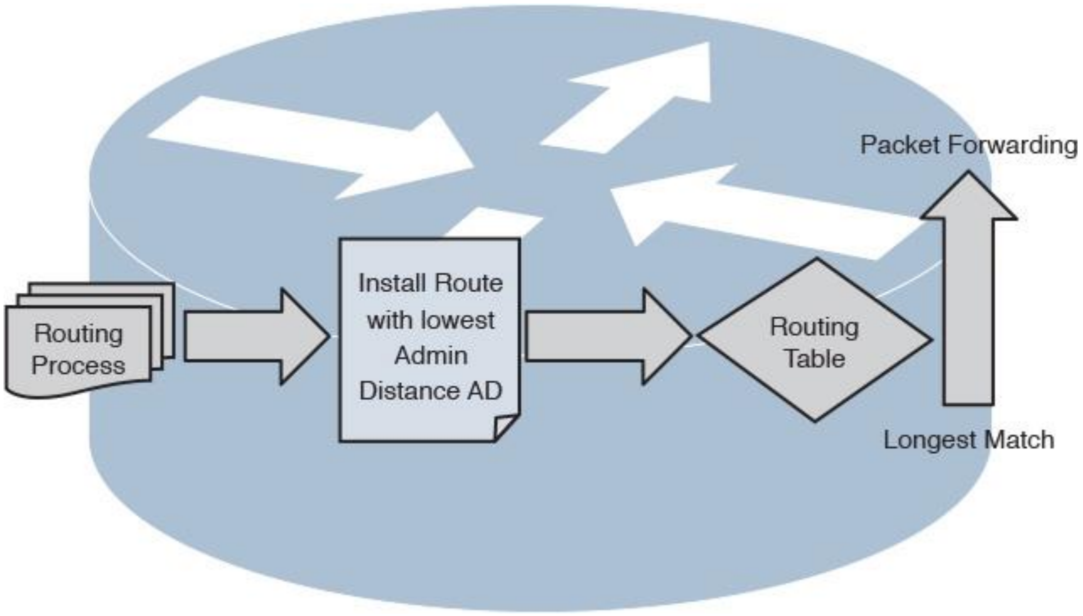

Figure 2-8 *Router's Forwarding Decision*

Link-State Routing Protocol Design Considerations

Open Shortest Path First (OSPF) and Intermediate System-to-Intermediate System (IS-IS) protocols as link-state routing protocols have a common conceptual characteristic in the way they build, interact, and handle L3 routing to some extent.Figure 2-9 illustrates the process of building and updating a link-state database (LSDB).

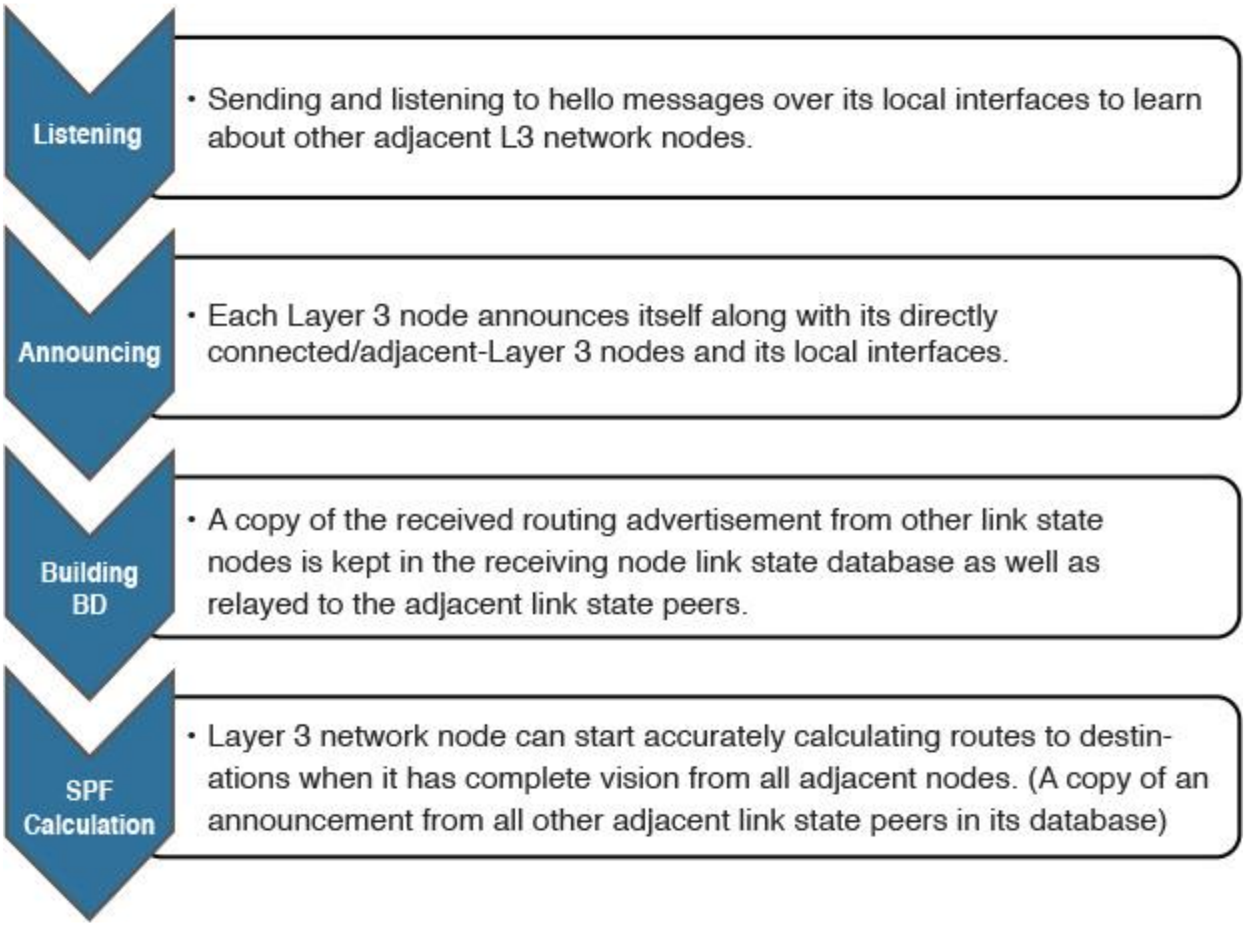

Figure 2-9 *Process of Building an LSDB*

It is important to remember that although OSPF and IS-IS as link-state routing protocols are highly similar in the way they build the LSDB and operate, they are not identical! This section discusses the implications of applying link-state routing protocols (OSPF and IS-IS) on different network topologies, along with different design considerations and recommendations.

Link-State over Hub-and-Spoke Topology

In general, some implications should be considered when link-state routing protocols are applied on a hub-and-spoke topology, including the following:

- There is a concern with regard to scaling to a large number of spokes, because each spoke node typically will receive all other spoke nodes' link-state information, because there is no effective means to control the distribution of routing information among these spokes.
- Special consideration must be taken to avoid suboptimal routing, in which traffic can use remote sites (spokes) as a transit site to reach the hub or other spokes.

For instance, summarization of routing flooding domains in a multi-area/flooding domain design with multiple border routers requires specific routing information between the border routers (Area Border Routers [ABRs] in OSPF or L1/L2 in IS-IS) over a nonsummarized link, to avoid using spoke sites as a transit path, as illustrated in Figure 2-10 [13].

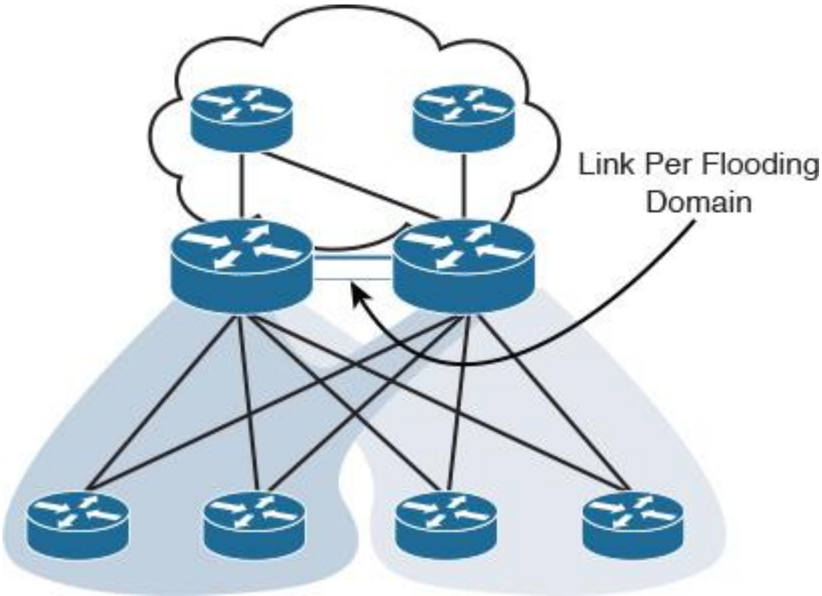

Figure 2-10 *Multi-Area Link State: Hub and Spoke*

So, for each hub-and-spoke flooding domain to be added to the hub routers, you need to consider an additional link between the hub routers in that domain. This is a typical use case scenario to avoid suboptimal routing with link-state routing protocols. However, when the number of flooding domains (for example, OSPF areas) increases, the number of VLANs, subinterfaces, or physical interfaces between the border routers will grow as well, which will result in scalability and complexity concerns. One of the possible solutions is to have a single link with adjacencies in multiple areas. (RFC 5185) [13]. For instance, in the scenario illustrated in Figure 2-11, there is a hub-and-spoke topology that uses OSPF multi-area design.

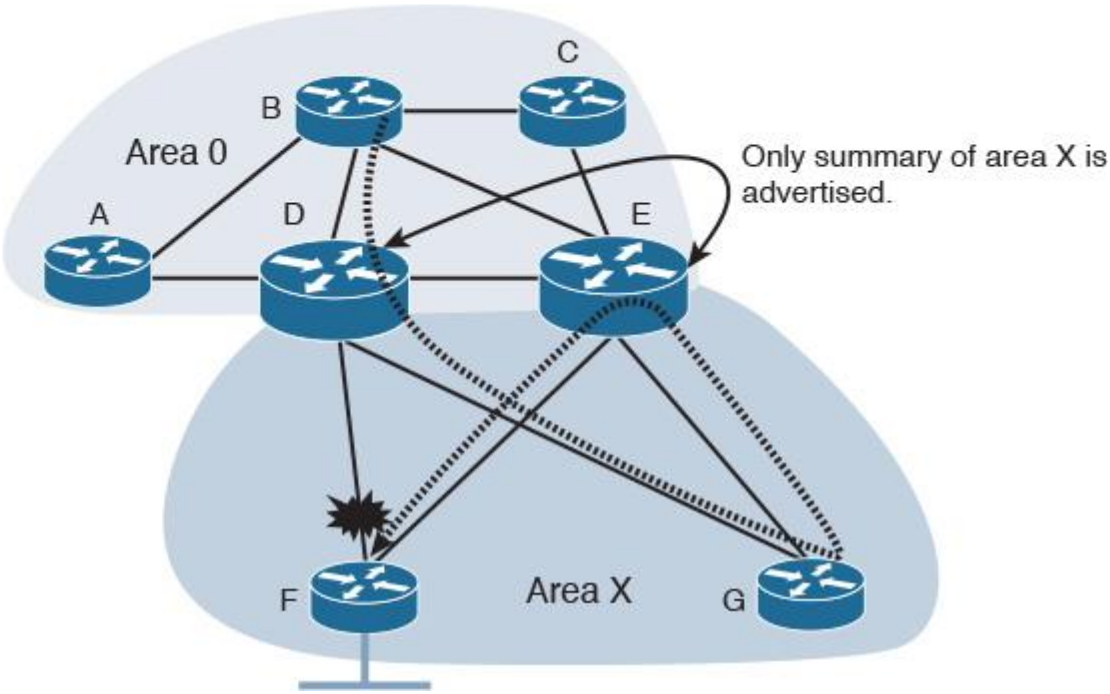

Figure 2-11 *Multi-Area OSPF: Hub and Spoke*

If the link between router D and router F (part of OSPF area X) fails, any traffic from router B destined to the LAN connected to router F going toward the summary advertised route by router D will traverse across the more specific route over the path G, E, then F.

To optimize this design during this failure scenario, there are multiple possible solutions, and here network designers must decide which solution is the most suitable one with regard to other design requirements such as application requirements where delay could affect critical business applications:

■ Place the inter-ABR link (D to E) in area X (simple and provide “north to south” optimal routing in this topology).

■ Place each spoke in its own area with link-state advertisement (LSA) type 3 filtering. (May lead to complex operations and limited scalability; “depends on the network size.”)

■ Disable route summarization at the ABRs; for example advertise more specific routes from ABR router E. (May not always be desirable because this means reduced scalability and the loss of some of the value of the OSPF multi-area design.)

Note

The link between the two hub nodes (for example, ABRs) will introduce the potential of a single point of failure to the design. Therefore, link redundancy (availability) between the ABRs may need to be considered.

If IS-IS is applied to the topology in Figure 2-11 instead, using a similar setup where IS-IS L2 is to be used instead of the area 0 and IS-IS L1 is to be used by the spokes, the simplest way to optimize this architecture is to put the links between the border routers in IS-IS L1-L2 (overlapping levels capability), where we can extend L1 to overlap with L2 on the border router (ABR in OSPF), as illustrated in Figure 2-12. This will result in a topology that can support summarization with more optimal routing with regard to the failure scenario discussed above.

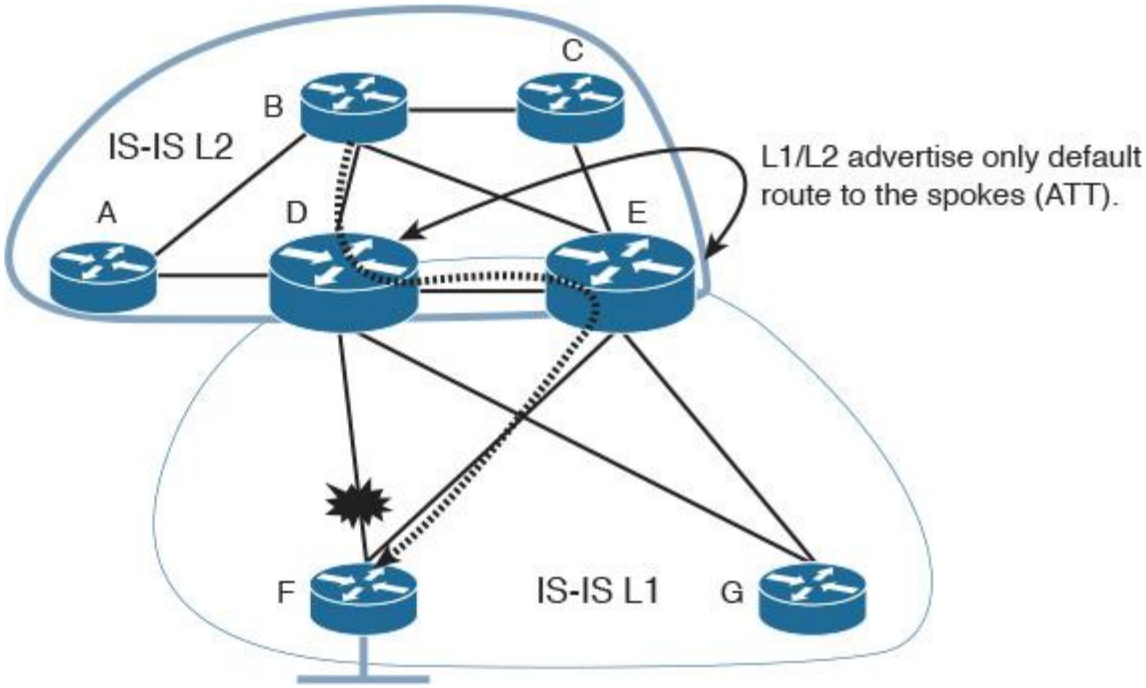

Figure 2-12 *Multilevel IS-IS: Hub and Spoke*

Note

OSPF is a more widely deployed and proven link-state routing protocol in enterprise networks compared to IS-IS, especially with regard to hub-and-spoke topologies. IS-IS has limitations when it works on nonbroadcast multiple access (NBMA) multipoint networks.

OSPF Interface Network Type Considerations in a Hub-and-Spoke Topology

Figure 2-13 summarizes the different possible types of OSPF interfaces in a hub-and-spoke topology over NBMA transport (typically either Frame Relay or ATM), along with the associated design advantages and implications of each [13].

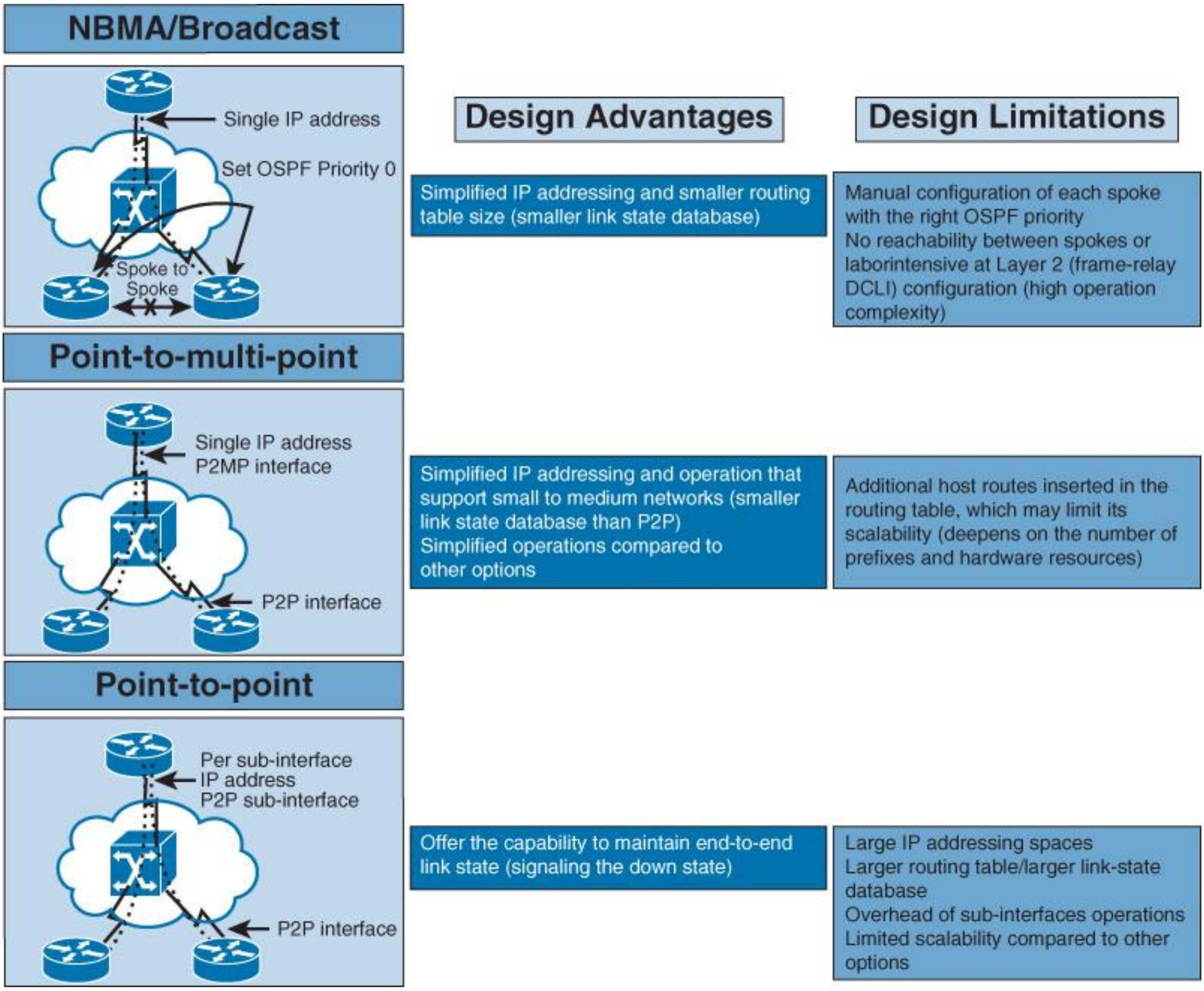

Figure 2-13 *OSPF Interface Types Comparison: Hub and Spoke*

Link-State over Full-Mesh Topology

Fully meshed networks can offer a high level of redundancy and the shortest paths. However, the substantial amount of routing information flooding across a fully meshed network is a significant concern. This concern stems from the fact that each router will receive at least one copy of every new piece of information from each neighbor on the full mesh. For example, in Figure 2-14, each router has four adjacencies. When a router's link connect to the LAN side fails, it must flood its LSA/LSP to each of the four neighbors. Each neighbor will then flood this LSA/LSP (link-state package) again to its neighbors. This process will culminate in a process like a broadcast being sent, due to this full-mesh connectivity and reflooding [13].

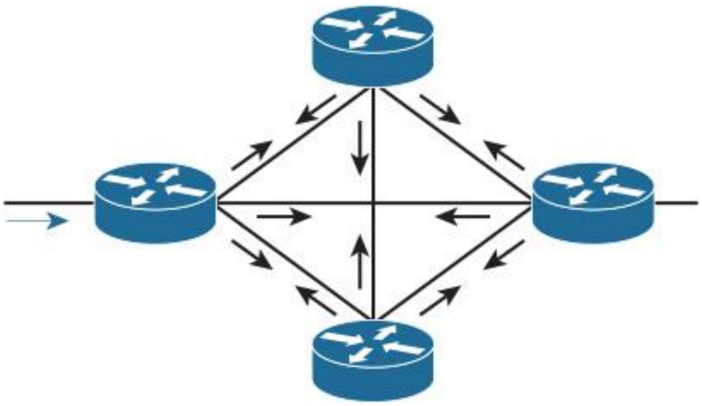

Figure 2-14 *Link-State: Full-Mesh Topology*

With link-state routing protocols, you can use the mesh group technique to reduce link-state information flooding in a full-meshed topology [23]. However, with link-state routing protocols in failure scenarios over a meshed topology, some routers may know about the failure before others within the mesh. This will typically lead to a temporarily inconsistent LSDB across the nodes within the network, which can result in transient forwarding loops. Even though the concept of a loop-free alternate (LFA) route can be considered to overcome situations like this, using LFA over a mesh topology will add complexity to the control plane.

Note

Later in this chapter, in the section "Hiding Topology and Reachability Information," more details are provided about flooding domain and route summarization design considerations for link-state routing protocols, which can reduce the level of control plane complexity and optimize link-state information flooding and performance.

Note

Other mechanisms help to optimize and reduce link-state LSA/LSP flooding by reducing the transmission of subsequent LSAs/LSPs, such as OSPF flood reduction (described in RFC 4136). This is done by eliminating the periodic refresh of unchanged LSAs, which can be useful in fully meshed topologies

OSPF Area Types

Table 2-3 contains a summarized review of the different types of OSPF areas [21, 22].

Area Type	Advertised Route
Stubby	All routes except type 5, external routing information
Totally stubby	Internal area routes + default route (both type 3 and 5 LSAs are suppressed.)
Not so stubby (NSSA)	All routes with the ability to inject/originate external routing information (type 7 LSA)
Totally NSSA	Internal area routes + default route, with the ability to inject/originate external routing information (type 7 LSA)

Table 2-3 *OSPF Area Types*

Each of the OSPF areas allows certain types of LSAs to be flooded, which can be used to optimize and control route propagation across OSPF routed domain. However, if OSPF areas are not properly designed and aligned with other requirements, such as application requirements, it can lead to serious issues because of the traffic black-holing and suboptimal routing that can appear as a result to this type of design. Subsequent sections in this book discuss these points in more detail.

Figure 2-15 shows a conceptual high-level view of the route propagation, along with the different OSPF LSAs, in an OSPF multi-area design with different area types.